



HBC Diabetes Diary 2026

v20.3 Type 1 Diabetes APP

User manual and installation guide

For people living with insulin-treated diabetes, their relatives and their treating physicians.

August 2026

Table of contents

Table of contents	2
1. Introduction: what this application is and who it is for.....	4
2. What is new in v20	5
3. Glossary: the terms used in this manual	6
4. The application view by view	8
4.0 Screenshots: what the application looks like.....	8
4.1 Overview (Dashboard)	11
4.2 Charts.....	11
4.3 Entries	12
4.4 Statistics	14
4.5 Foods	16
4.6 Sync.....	16
4.7 Settings	18
4.8 New entry	23
4.9 Quick reference: buttons, icons, indicators	24
5. Diabetes, insulin and nutrition — what's worth knowing.....	26
5.1 Type 1 diabetes in brief	26
5.2 How insulin works — the basal-bolus principle	26
5.3 Insulin injection technique, step by step	26
5.4 Hypoglycemia and hyperglycemia — recognition and response	27
5.5 Carbohydrate counting basics.....	28
5.6 The effect of exercise on blood glucose	28
5.7 “Sick day” rules (fever, vomiting).....	29
6. Bolus calculation in detail	29
6.1 The formula in one place	29
6.2 Carb-coverage dose (ICR).....	30
6.3 Correction dose (sensitivity factor, ISF)	30
6.4 The role of active insulin (IOB)	30
6.5 One complete worked example	30
6.6 When and why the suggestion can be overridden.....	31
7. A typical day with the application — a practical example.....	31
8. The phases of installation	32
8.1 First phase: publishing on GitHub Pages	33

8.2 Installing on an Android phone	35
8.3 Installing on an iPhone	35
8.4 Installing on a Windows or Mac computer	35
8.5 Taking over the data of an earlier (v6, v7 or v8) diary.....	35
8.6 First settings, together with the treating physician	36
8.7 Updates in the future	36
9. Google Drive sync: the follower sees the values live	37
9.1 How it works	37
9.2 One-time Google setup (about 20 minutes, free)	37
9.3 Setting up the owner (owner mode)	40
9.4 Sharing the folder and setting up the follower (follower mode)	40
9.5 Backup in practice: the monthly routine	41
10. Loading CGM data (for sensor users)	41
11. Publishing in the Google Play store.....	42
11.1 Creating a developer account	42
11.2 Building the Android package	42
11.3 Upload and store listing	42
12. Troubleshooting: frequent questions.....	43
13. Medical and privacy notes	44
13.1 From a specialist's point of view	44
13.2 Where the data is	44
13.3 Summary	44
14. The SOS emergency feature	45
14.1 Setting it up	45
14.2 Opening the SOS page	45
14.3 Automatic alert on a low reading	46
14.4 Important limitations and advice	46

1. Introduction: what this application is and who it is for

The HBC Diabetes Diary is a personal health diary for people with insulin-treated diabetes. It runs on phones, tablets, laptops and desktop computers, works without an internet connection, and all data remains on the user's own device. The diary records blood glucose readings, the carbohydrate content of meals and the insulin administered, then turns them into clear charts and statistics.

The application provides five key services that a paper diary cannot. First: it performs calculations for the user — for example, it suggests a pre-meal insulin dose based on values agreed with the treating physician. Second: it gives immediate feedback — for example, it issues a warning when a low blood glucose value is recorded. Third: it can be shared, so a designated follower — a relative or the treating physician — can view fresh readings live on their own phone. Fourth: it is bilingual (Hungarian and English), and the blood glucose unit can be switched as well. Fifth — new in v9.0 — it produces a printable medical report for any freely chosen date range, which can be taken to an appointment as a PDF.

What the application does not do

The application is not a medical device and does not replace medical advice. All of its calculations — including the insulin dose suggestion and the estimated HbA1c and GMI values — are for information only. Dose decisions must in every case follow the guidance of the treating physician. The application displays this warning in the appropriate places as well.

Important medical warning

In case of feeling unwell, first measure blood glucose and act according to the learned protocol; attend to logging only afterwards. Example: at a reading of 3.2 mmol/l, first take 15 grams of fast-acting carbohydrate (3 glucose tablets or 150 ml of fruit juice), re-check after 15 minutes, and record the value only after that.

2. What is new in v20

This chapter briefly summarizes what changed compared to v19, then gives a short, topic-by-topic overview of earlier developments as well (v8–v19) — instead of a full version-by-version technical changelog, it focuses on what matters for day-to-day use. Automatic temperature lookup can be switched off individually in Settings; the other new features are automatically available to every user.

- **Automatic temperature lookup (v20.1):** when the Activity/heat profile block appears, the temperature is now looked up automatically by default, based on the device's location; it can be switched off in Settings, which restores the previous manual “🔍 Look up” button. (4.7, 4.8)
- **Foods in the entry — Amount and carbs/100g reworked (v20.1–v20.2):** fields for selected foods: Time | Mode (g/ml/pcs) | Amount | Carbs/100g (or carbs/100ml) | Delete — the item's final carb content is a calculated, displayed value; the detected Mode can always be overridden manually. (4.5, 4.8)
- **Activity picker — collapsed by default (v20.1):** the searchable “Activity (choose from earlier entries)” list now appears collapsed by default and opens on tap. (4.8)
- **Active basal insulin — Lantus and similar (v20.1):** a new, informational-only estimate of active basal insulin, based on manufacturer effect-duration data; it is NOT subtracted from the bolus suggestion. (4.1, 4.4)
- **Dawn phenomenon / Somogyi effect detection (v20.1, informational):** the diary automatically recognizes and distinguishes the two typical patterns behind an unexplained morning glucose rise — pattern recognition only, not a diagnosis. (4.4)
- **Responsive check — mobile and desktop views (v20.2):** after a thorough review, fixed the mobile layout of “Foods in the entry” and improved navigation on both mobile and desktop.

Earlier versions (v8–v19) — key developments, summarized

The diary has been evolving continuously since 2019. The topic-based summary below gathers the key developments of the earlier major versions (v8–v19) — showing which areas grew, rather than listing every point release in detail.

- **Bolus calculator and activity/heat profiles (v14–v19):** freely editable, percentage-based correction presets for physical activity and heat, with temperature lookup and a calibrated, retrospective analysis on the Statistics page.
- **More precise data entry (v12–v20):** a separately recorded insulin injection time, freely entered grams and carbs per food item, a weight field, and typo protection for unusual glucose values.
- **Clearer user interface (v8–v19):** unified “Today's entries” layout, searchable activity picker, profile photo, a custom icon set, custom color and background themes, and a bottom mobile navigation bar.
- **Statistics and medical report (v8–v18):** GMI indicator and daily-pattern chart, an emailable or downloadable PDF report with official header data, and dawn-phenomenon detection.
- **Sync, roles and safety (v9–v14):** Google Drive sync with automatic sign-in, Owner/Follower roles with private entries, and the SOS emergency feature with automatic alerting.
- **Foundations (v8 and earlier):** today's interface, the entry types and the food database were established, along with full Hungarian and English localization.

The full, itemized changelog of older versions is available from the developer on request; this manual intentionally focuses on the developments that matter for everyday use.

3. Glossary: the terms used in this manual

The table explains every term that appears in this manual or in the application. When encountering an unfamiliar word, its meaning can be looked up here quickly.

Term	Plain-language meaning
Blood glucose (BG)	The sugar level of the blood. In Hungary it is measured in mmol/l, e.g. 5.6 mmol/l. In the American unit, mg/dl, the same value is 101 mg/dl.
mmol/l and mg/dl	Two units for the same value. Conversion: 1 mmol/l = 18 mg/dl. It can be switched on the Settings tab.
Carbs (CH)	The blood-glucose-raising component of food, measured in grams. Example: a bread roll is about 30 grams of carbs.
Bolus	The rapid-acting insulin taken with a meal, e.g. Humalog. It covers the carbohydrate content of the food.
Basal	The once-daily long-acting insulin, e.g. Lantus. It works steadily in the background all day.
ICR (carb-to-insulin ratio) Insulin-to-Carbohydrate Ratio (ICR)	Indicates how many grams of carbohydrate 1 unit of rapid insulin covers. Example: with an ICR of 10, a lunch containing 60 grams of carbs requires 6 units of insulin. The value may differ by time of day.
Sensitivity (ISF) Insulin Sensitivity Factor	Indicates how much 1 unit of rapid insulin lowers blood glucose. Example: with a sensitivity of 2.5 mmol/l, a reading of 9.0 mmol/l is expected to drop to 6.5 mmol/l after 1 unit.
IOB (insulin on board) Insulin On Board	The amount of previously injected insulin still active. Example: 2 hours after injecting 4 units, about 2.3 units are still working. The application subtracts this from the next suggestion, protecting against stacked doses.
TIR (time in range) Time in Range	The percentage of readings that fell within the target band (by default between 3.9 and 10.0 mmol/l). Example: 78% TIR means almost eight out of ten readings were in range. Medical guidance generally considers above 70% good.
HbA1c	The proportion of sugar bound to red blood cells, reflecting the average blood glucose of the past 2–3 months. The application computes an estimate from the readings; the precise value comes only from a laboratory test.
GMI Glucose Management Indicator	Glucose Management Indicator: an HbA1c-like percentage computed from average glucose with an international formula. GMI shows how your sugar metabolism has been over the past few days (usually requires at least 14 days of CGM data) A common reference number in the CGM world — also an estimate, not a laboratory result.
SD and CV SD - Standard Deviation	Variability measures in the medical report. SD (standard deviation) shows how far readings scatter around the average; CV is the same as a percentage — below 36% is generally considered stable.

Term	Plain-language meaning
CGM Continuous Glucose Monitoring	Continuous glucose monitor sensor (e.g. FreeStyle Libre or Dexcom) measuring every minute or every 5 minutes. The sensor's factory export file can be imported into the application.
PWA Progressive Web App (PWA)	Progressive web application: a website that can be installed on phones and computers like an app, with an icon, full screen and offline operation.
Google Drive	Google's free cloud storage (15 GB with every Google account). The application saves a copy of the diary there when sync is enabled.
Sync	Automatic reconciliation of data between two devices. Here: uploading the diary to Drive and downloading fresh values to the follower's phone.
Owner and Follower	The two roles of sharing. The Owner is the user keeping the diary; the Follower is the person — a relative or the treating physician — who can view the shared diary live, in read-only mode.
Hypoglycaemia (hypo)	Blood glucose lower than normal — in the application, a value below the configured lower limit (3.9 mmol/l by default). Symptoms may include trembling, sweating, weakness. Action: 15 grams of fast carbs, re-check after 15 minutes.
Hyperglycaemia (hyper)	Blood glucose higher than normal — in the application, a value above the upper limit (10.0 mmol/l by default). The application marks it red on the charts and adds a correction to the bolus suggestion.
JSON and CSV	Two file formats. JSON is the complete, restorable backup of the diary. CSV is a table that Excel opens, for medical review or personal analysis.
Active basal insulin (basal IOB)	An estimate of how much of the previously injected long-acting (basal) insulin is still active, based on the manufacturer's official duration-of-action data. Unlike rapid-insulin IOB, this is purely informational — it is not subtracted from the bolus suggestion.
Dawn phenomenon	A morning blood-glucose rise, typically without carb intake or rapid insulin, caused by the body's early-morning hormonal processes. The diary detects and flags this automatically — for information only, not a diagnosis.
Somogyi effect	A morning blood-glucose rise similar to the dawn phenomenon, but preceded by a low blood glucose (hypoglycaemia) overnight — the body's rebound response. The diary flags this separately from the dawn phenomenon.
Carbs/100g and carbs/100ml	The carbohydrate content of a food or liquid per 100 grams or per 100 millilitres — typically how it appears on a product's nutrition label. The diary uses this together with the amount actually consumed to calculate the item's final carb (g) value.

4. The application view by view

On a phone, a quick bar at the bottom of the screen holds the five most-used pages (Overview, Charts, New entry, Entries, Statistics), while the  hamburger menu in the top right corner of the header reaches every page — including Foods, Sync and Settings. On desktop computers and tablets all seven tabs are visible in a row under the header on every page, and the new-entry button sits in the top right corner of the header. This chapter introduces each view: its purpose, its contents, and its recommended use.

4.0 Screenshots: what the application looks like

The following screenshots show the key elements using the application's real colors, logo and labels, so that using it feels familiar from the very first moment.

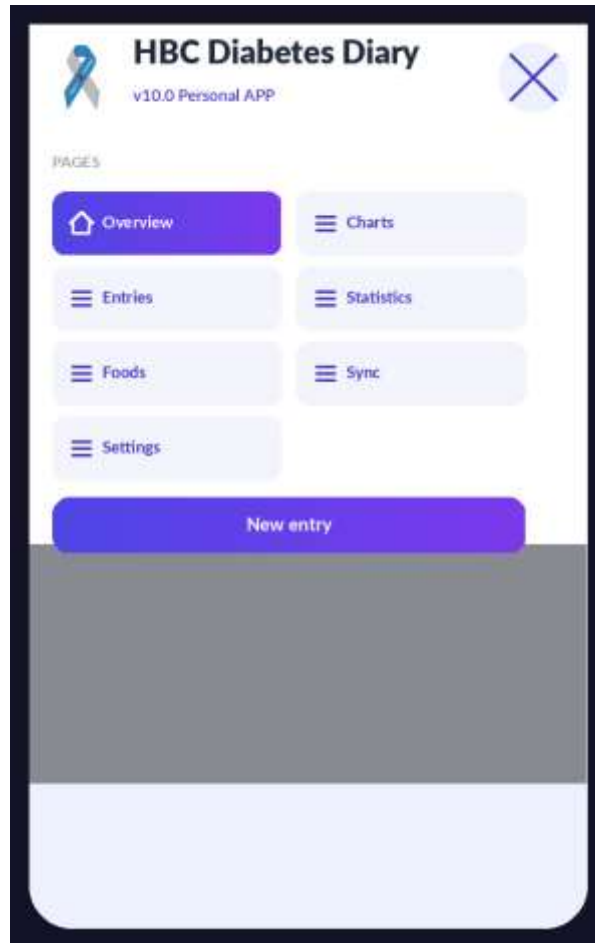
a) Desktop header + top button bar (v10 — larger, evenly split)



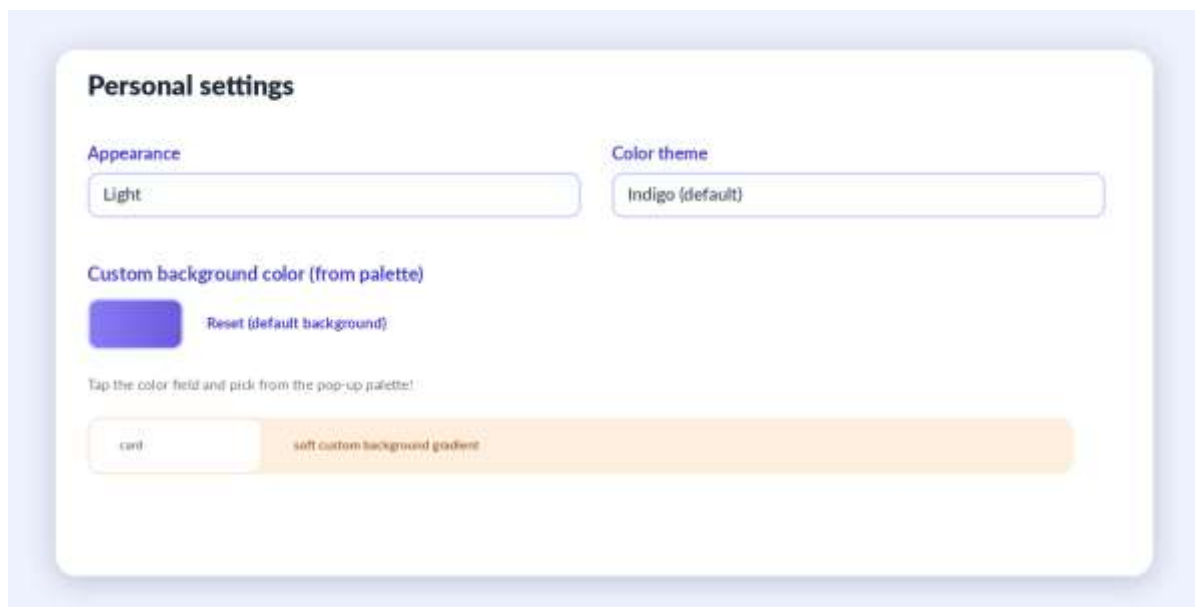
b) Mobile bottom bar + hamburger button



c) Opened hamburger menu (v10 — opens BELOW the header)



d) Appearance / Color theme setting — custom background color picker



e) New entry form (date / time field)

The 'New entry' form is a white card with rounded corners and a purple shadow. It features a purple header bar with the text 'New entry'. Below the header, there are four input fields with labels: 'Time (can be changed later!)', 'Meal type', 'Blood glucose (mmol/l)', and 'Carbohydrate (g)'. The first field contains the text '2026-07-14 16:05', the second contains 'Lunch', the third contains '9.0', and the fourth contains '60'. At the bottom of the card is a purple button labeled 'Save'.

f) Entry cards with indicator colors (green / amber / red)

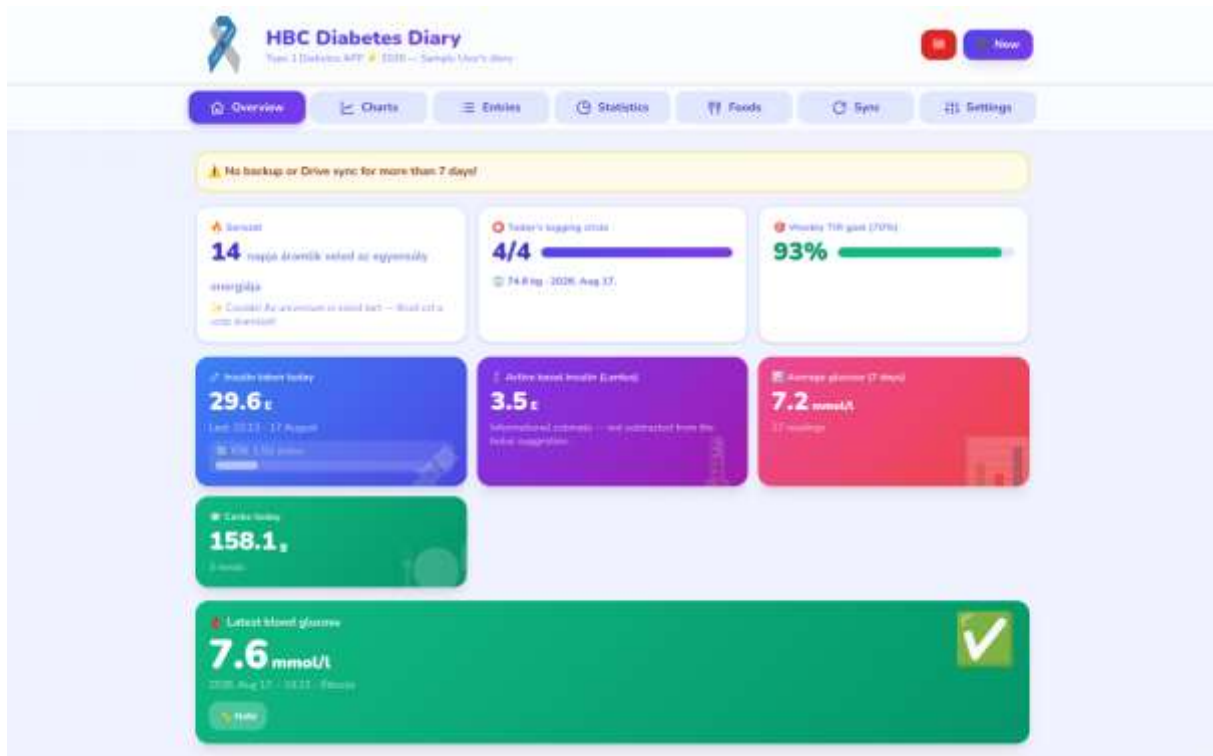
The screenshot shows three entry cards stacked vertically. Each card has a colored vertical bar on the left side: green for the first card, orange for the second, and red for the third. The cards contain the following information:

- 07:12** (Green bar): 6.2 mmol/l - Breakfast - 42 g carbs - 4.5U Humalog
- 12:41** (Orange bar): 9.0 mmol/l - Lunch - 60 g carbs - 6.0U Humalog
- 16:05** (Red bar): 3.4 mmol/l - Check - Hypo treated

Each card has a pencil icon and a trash can icon on the right side. At the bottom of the screen, there is a legend with three colored circles and their corresponding labels: a green circle for 'In range', an orange circle for 'High', and a red circle for 'Low'.

4.1 Overview (Dashboard)

This is the home screen, the summary of the day. At the top there are three motivation tiles: Streak counts how many days the diary has been kept complete (e.g. 12 days of complete logging), Today's logging circle shows how many of the 4 daily entries are done (e.g. 2/4), and the Weekly TIR goal shows time in range against the 70% target. Below them are three coloured summary tiles: insulin taken today, the 7-day glucose average and today's carbs. The largest card shows the latest blood glucose value: green if in range, amber if low, red if high. A short note for the physician can also be added here, for example: woke at 3 am, sweating. In the evening the application indicates if no basal insulin has been logged that day.



4.1 Overview — the home screen (illustrated with synthetic sample data).

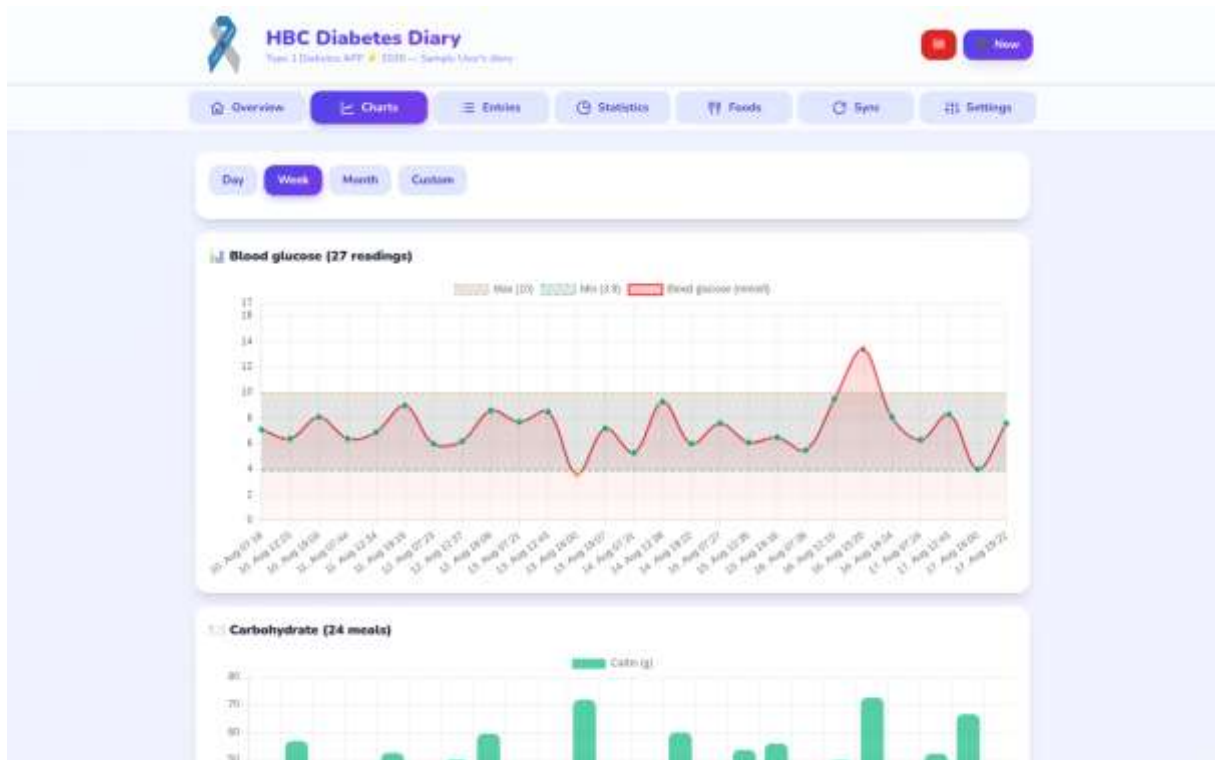
New in v18: a progress bar below the insulin-on-board (IOB) figure now also shows the active amount; the post-meal measurement reminder signals in a green banner after the configured time (default: 100 minutes) that it is time for a control measurement — acknowledge it with the “OK” button; in follower mode a short weekly summary card (TIR, average, number of hypos) appears once a week.

New in v18.5: tapping the logo at the top left of the header (or Tab + Enter/space on keyboard) instantly switches between dark and light view with no confirmation prompt; the setting also updates the Settings → Appearance field and is saved immediately.

New in v20.1: the estimated active amount of basal insulin (Lantus and similar) now appears on its own stat card, and also on a combined line chart with the rapid insulin — purely informational, not subtracted from the bolus suggestion.

4.2 Charts

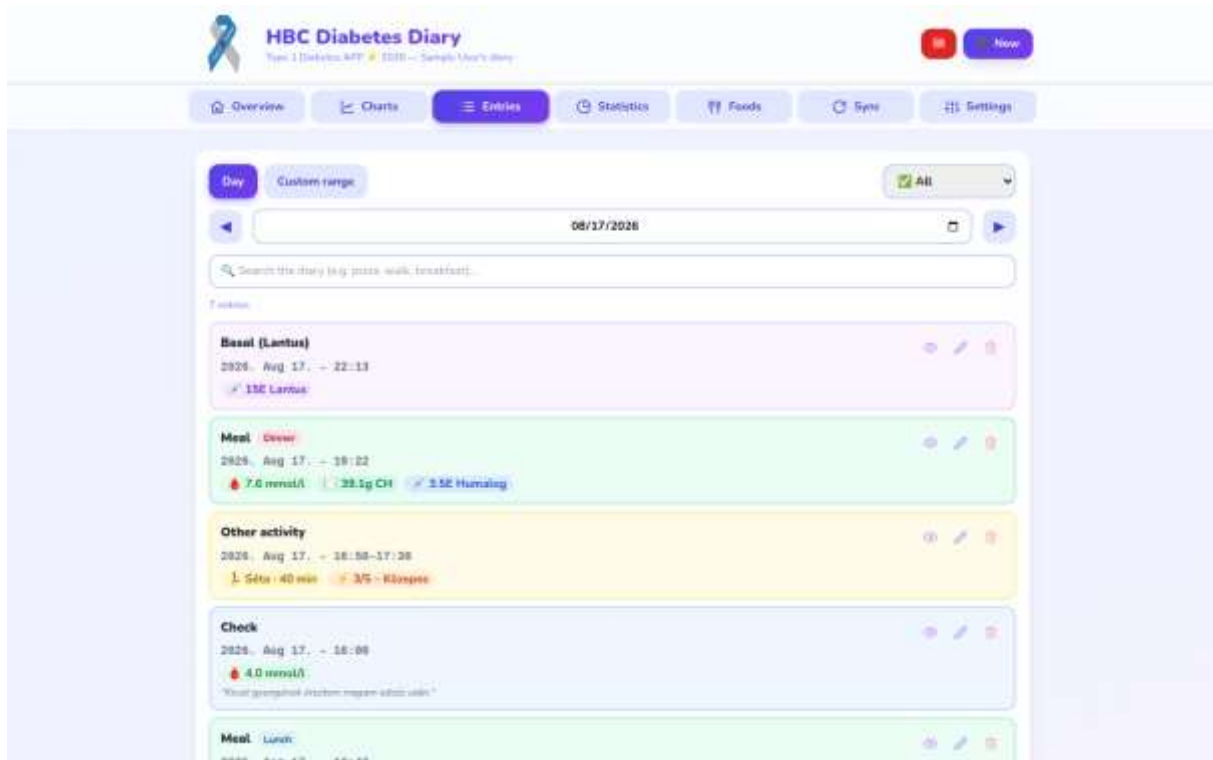
Three charts show the recent period: blood glucose (line chart), carbohydrate and insulin (bar charts). The range is switched with buttons: Day, Week, Month, or a Custom period where the start and end dates can be set freely (from-to). On the glucose curve a green band marks the target range, and the colour of each point grades the reading: green in range, amber low, red high. Imported CGM data appears on a separate chart. A practical example: before an appointment, set the range to Month and review the behaviour of the early-morning hours together with the physician.



4.2 Charts — glucose and carbohydrate curves (illustrated with synthetic sample data).

4.3 Entries

The list of all records in chronological order, filtered by day or by any from–to period, and also by type (Meal, Check, Basal, Other activity). Every card shows the time, blood glucose, carbs, insulin and note. With the pencil icon any entry can be corrected later; the bin icon deletes it. Example: if it turns out in the evening that 54 grams were recorded at lunch instead of 45, it can be corrected here with two taps.



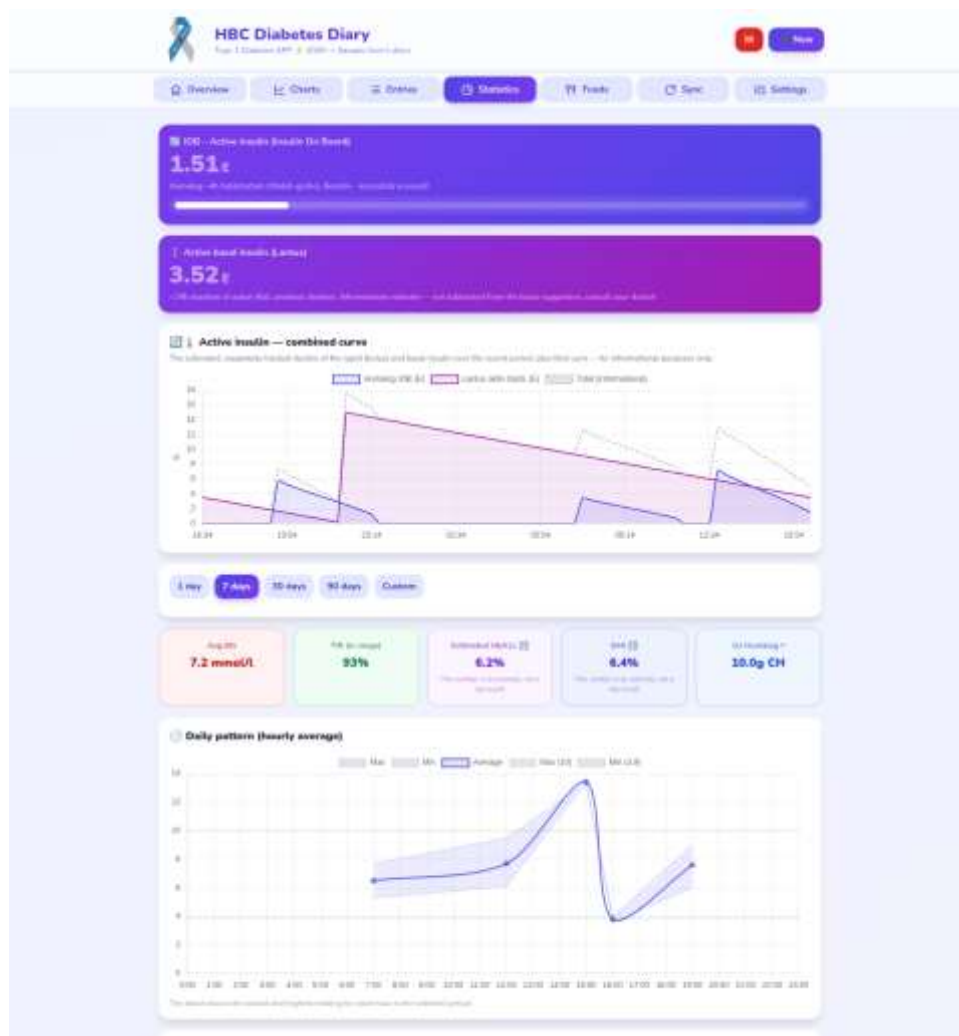
4.3 Entries — the diary's list view (illustrated with synthetic sample data).

New in v14: the entry card shows the duration next to the activity name and the physical activity level (1–5) on its own label; private entries carry a “Private” label (visible to the Owner only). Selecting the “Other activity” filter reveals a second dropdown that narrows the list to a specific activity name. In follower mode an eye icon replaces the pencil and the bin: tapping it opens every detail of the entry in a read-only view.

New in v18: free-text search — type for example “pizza” and the search brings up similar earlier entries from the whole diary (regardless of the date filter), with their dates.

4.4 Statistics

A numeric summary for the chosen period, set with buttons (1, 7, 30, 90 days) or with a custom from–to date range: average glucose, TIR percentage, estimated HbA1c, GMI, and the distribution of readings (low, normal, high). A separate card shows the currently active insulin (IOB) and the actual carb-to-insulin ratio computed from the diary. Next to the estimated HbA1c and the GMI the application notes: this number is an estimate, not a laboratory result. An example of interpretation: a 90-day TIR of 82% with an estimated HbA1c of 6.8% indicates good balance, to be confirmed by the next laboratory test. New in v9.0 is the Daily pattern chart: it groups the readings of the selected period by hour of day and shows the hourly average together with the band of the lowest and highest values. At a glance it reveals in which parts of the day blood glucose tends to rise or fall — for example the dawn phenomenon or a regularly under-estimated dinner.



4.4 Statistics — active insulin, key figures and daily pattern (illustrated with synthetic sample data).

New in v18: AGP view (percentile bands) — the international-standard chart: the thick line is the hourly median, the darker band is the middle 50% of readings (25th–75th percentile), the lighter one the middle 90% (5th–95th percentile). Fingerstick readings and loaded CGM data are combined. The view can be switched in Settings.

Creating the medical report (PDF)

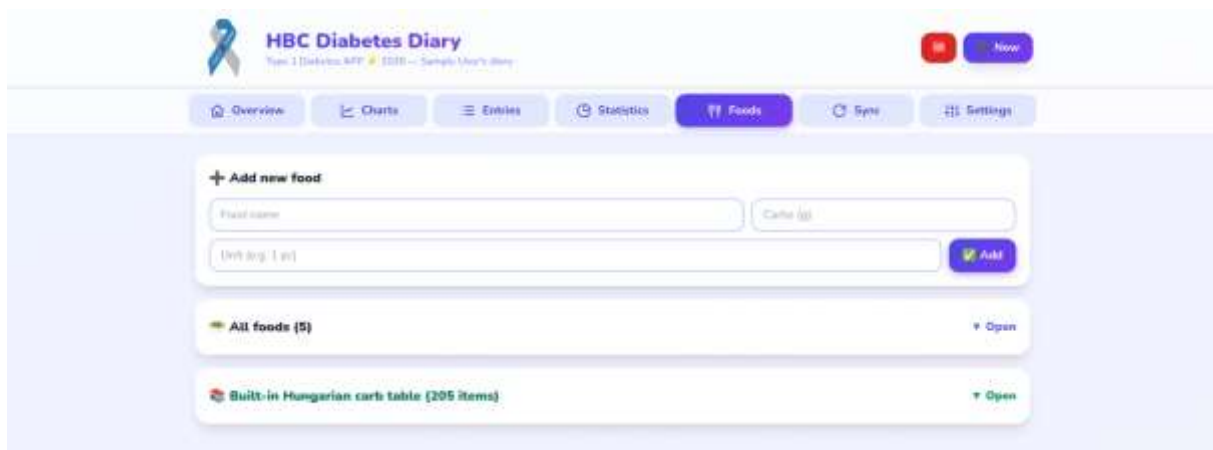
The Medical report card is located at the bottom of the Statistics tab. Set the start and end dates (from-to), then press Create report. A print-ready summary opens on a new page: the nickname (if configured) and the period in the header, followed by the key figures (number of readings, average, SD/CV, TIR, estimated HbA1c, GMI, low and high events, daily average insulin and carbs), the glucose curve of the whole period, the daily pattern, and finally the detailed log table. Use the Print button on the opened page to print on paper, or choose Save as PDF in the print dialog to create a file that can be sent by e-mail. The report is generated in the application's language and with the configured unit.

New in v15/v16: the Create report button opens a panel from which the report can also be sent directly by e-mail to your doctor from your own Gmail account. The recipient can be picked from the saved address list; the list is unlimited and editable. The message contains the formal cover letter (the patient's name, phone number and e-mail address), while the report itself arrives as an attached, printable PDF file that opens in any PDF viewer. The familiar PDF / Print option is available via the PDF / Print button of the panel.

New in v20.1: the estimated active basal insulin appears on a combined line chart with the rapid insulin; a new "Dawn phenomenon / Somogyi effect" summary shows if the diary detects a notable morning blood-glucose rise without carb intake — both purely informational, and both are included in the medical report.

4.5 Foods

The user's own food database. It starts with 22 foods and their carb values, for example: a 1 cm slice of sourdough loaf is 15 grams of carbs, a medium apple is 15 grams. It can be extended at any time by entering a name, carb value and unit (e.g. cottage-cheese pancake, 25 grams, 1 pc). Foods added here appear in the quick picker of the New entry screen, so at mealtime they can be added with a single tap.



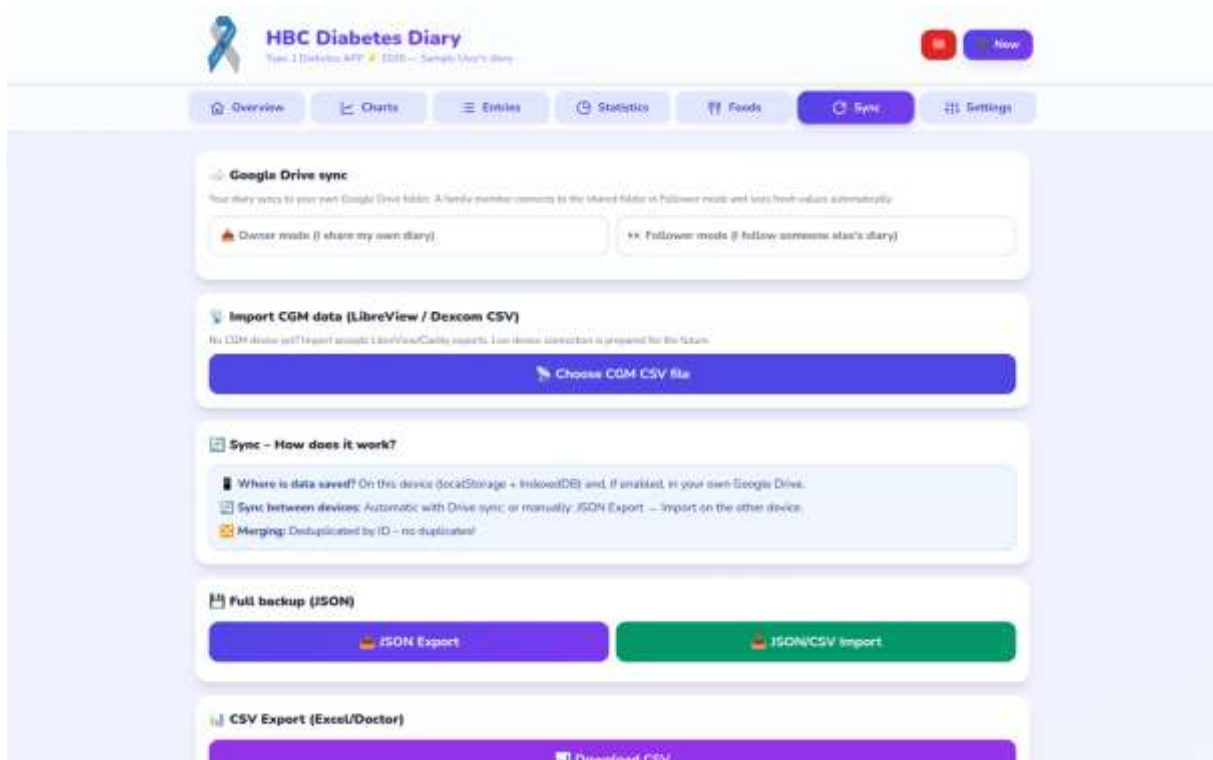
4.5 Foods — your own food database (illustrated with synthetic sample data).

New in v18: built-in Hungarian carb table — the food search of the New entry form also searches a built-in list of hundreds of Hungarian foods (typical portion + carbs) in addition to your own foods; on the Foods page the full 205-item list can be browsed by category, and any item can be copied into your own foods with the **+** button. The values are average, rounded figures; exact carbs may differ by product.

New in v20.1–v20.2: the food input fields switched to Amount (g) + Carbs/100g (or, for liquids, Amount (ml) + Carbs/100ml). The app detects whether an item is weight-, liquid- or piece/serving-based; this can always be overridden with the Mode dropdown. For piece/serving-based foods (e.g. “1 pcs”, “1 serving”, “1 tablespoon”) the Amount field is the piece count, and behaves just like the earlier $\times 1/\times 2/\times 3$ quick-add multiplier.

4.6 Sync

All data backup and sharing features are located here: setting up Google Drive sync (details in chapter 9), importing a CGM file (chapter 10), full backup to a JSON file, CSV download for Excel or for the physician, and text copy-paste between two devices. JSON is the complete, restorable copy of the diary; CSV is a tabular format that Excel opens. Recommendation: once a month press JSON Export and store the file — that is the backup. If the latest backup is older than 7 days, the Overview screen issues a warning.

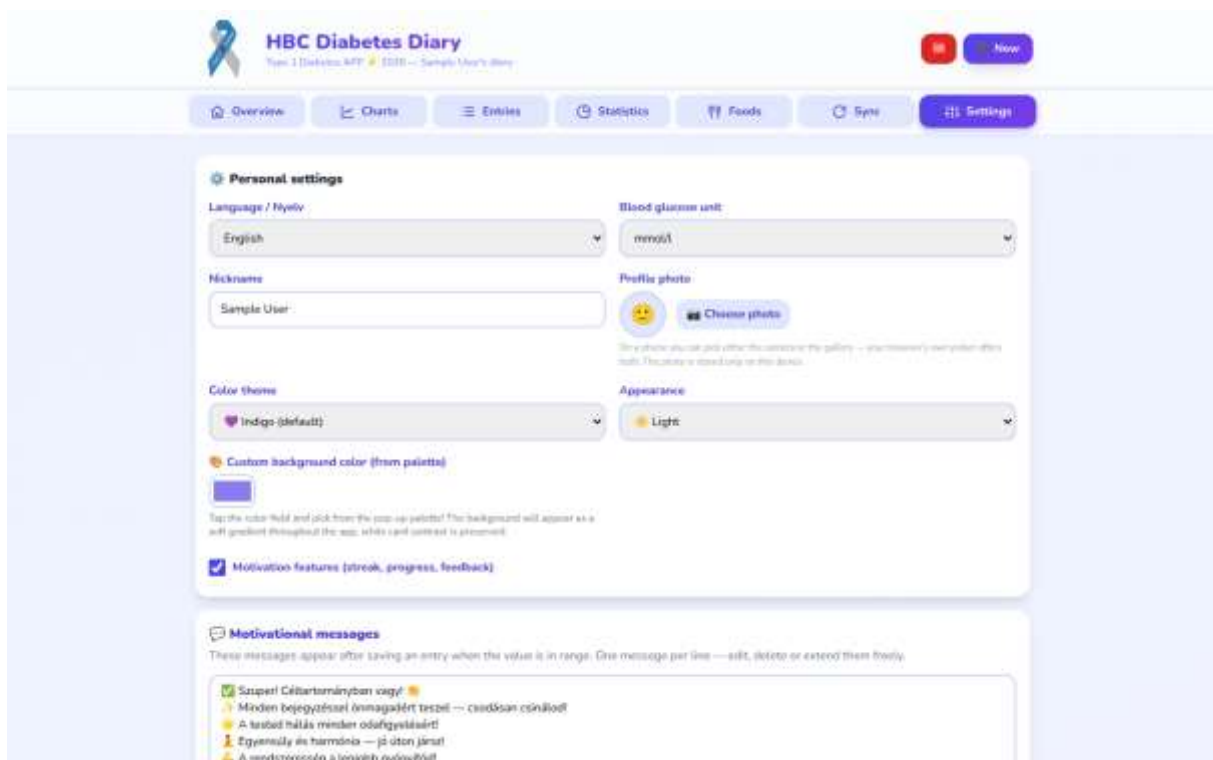


4.6 Sync — backup, import/export and CGM import (illustrated with synthetic sample data).

New in v14: the Google Drive sync settings now include an “Automatic sign-in — Google account (e-mail)” field (in owner and follower mode alike). With the account set, the app signs in by itself, WITHOUT the account picker list — after a one-time manual authorisation per device. This matters most in an emergency: an unfamiliar helper does not have to pick the right account from the long list of Google accounts on the device.

4.7 Settings

The personal settings come in four groups. The first group is appearance: language (Hungarian or English), blood glucose unit (mmol/l or mg/dl), nickname, colour theme (indigo, teal, rose — and from v9.0 also a custom color chosen from a pop-up color palette), light, dark or automatic mode, and the switch for motivation features. The second group is the blood glucose thresholds: low limit, high limit and target value. The third group — new in v9.0 — is the time-of-day boundaries: here it can be set until what hour Breakfast, Mid-morning snack, Lunch, Afternoon snack and Dinner last (this drives the automatic meal-type suggestion), and until when the bolus calculator uses the morning and the midday ICR. The fourth group is insulin: the names of the rapid (bolus) and basal insulin — selectable from v9.0 in a drop-down list of the products available in Hungary, with a custom name still possible —, the action time in hours, the sensitivity, and the carb-to-insulin ratio (ICR) by time of day. The table in chapter 8.6 assists in completing the values together with the treating physician. From v10.0 a custom background color can also be chosen from a pop-up color palette: the selected color appears as a soft background gradient throughout the application, in both light and dark mode, while card contrast is preserved; it can be cleared at any time with the Reset button.



4.7 Settings — personal settings card (illustrated with synthetic sample data).

New in v18: Motivational messages card — the texts shown after saving an entry can be freely edited and extended, and the display duration can be set between 1 and 10 seconds; the texts of the Streak card can be rephrased; every new feature can be switched on/off on the New features (v19) card (enabled by default); a large font mode can be switched on for older or visually impaired users.

Below, EVERY card and field of the Settings page is described in detail, card by card — together with the default values. The “Save settings” button at the bottom of the cards stores every change.

Personal settings card

Personalises the language and appearance of the application.

- **Nyelv / Language:** Hungarian or English interface with a single tap. Stored data is language-independent — nothing is lost when switching, and you can switch back at any time.
- **Blood glucose unit:** mmol/l (Hungarian custom) or mg/dl. The app always stores values in mmol/l internally, so switching units is safe and reversible; every chart and statistic converts instantly.
- **Nickname:** appears in the header (“v19 Type 1 Diabetes APP  — Nickname’s diary”), and the Follower sees it too — so the relative always knows whose data they are looking at.
- **Colour theme + custom accent colour:** choose from preset themes, or pick your own accent colour on the pop-up palette (colour of buttons, bars, active elements).
- **Appearance:**  Light, Dark or Automatic — the latter follows the phone’s system setting (e.g. switches to dark in the evening by itself).
- **Quick switch with the header logo (v18.5):** besides the Appearance dropdown above, tapping the logo at the top left of the header also instantly switches between dark and light view, without confirmation — handy when you just need a quick switch (e.g. for evening reading). Starting from Automatic mode, the tap switches to the opposite of the current view, and from then on it lives on as a manual setting until you set the dropdown back to Automatic.
- **Custom background colour:** the colour picked from the palette appears as a soft gradient across the whole app, in light and dark mode alike; the “ Reset” button removes it.
- **Motivational elements (toggle):** switches the Streak / Daily logging circle / Weekly TIR goal cards of the Overview and the positive feedback after saving on or off. Enabled by default.

Motivational messages card (v18)

Full customisation of the positive messages that pop up after saving an entry.

- **Message list:** one message per line; edit, delete or extend them freely. When an in-range value is saved, the app picks one at random. If you delete every line, the factory defaults are used.
-  **Restore default messages:** brings back the six factory messages with one tap.
- **Display duration:** adjustable with a slider between 1 and 10 seconds so it can be read comfortably. Default: 3 s.

Streak card texts (v18)

The three texts of the Streak card of the Overview can be rephrased to your own taste.

- **Card title:** default “Streak”.
- **Text after the number:** default “days the energy of balance has been flowing with you”.
- **Encouraging message (3+ days):** appears from a 3-day streak; default “ Wonderful! The universe is with you — guard this beautiful flow!”.
-  **Restore default texts:** you can return to the factory texts at any time.

New features (v19) card

Every new v19 feature can be switched here separately; all are enabled by default (the only exception is large font mode). Enabled features are marked with a green frame.

- **Built-in Hungarian carb table:** 205 Hungarian foods with category buttons and search, in the quick picker of the New entry form and on the Foods page, fully offline.
- **AGP view (percentile bands):** the international-standard chart on the Statistics page — hourly median + 50% and 90% bands.
- **Fatty/protein-rich meal flag:** a toggle on the meal entry + warning about a delayed glucose rise.
- **Bluetooth glucose meter connection:** reads the last measurement from a standard Bluetooth meter. Requires Android + Chrome; not supported on iPhone.
- **Predicted low alert:** the Follower is alerted before the CGM value drops below the limit (trend-based, about 20 minutes ahead).
- **Post-meal measurement reminder + “Remind after meal” field:** adjustable between 30 and 240 minutes (default: 100; recommended 90–120); the Overview signals this long after a meal.
- **Diary search:** free-text search on the Entries page — across the whole diary, regardless of the date filter.
- **Weekly summary for the follower:** once a week a short digest card (TIR, average, number of hypos) on the Follower device.
- **Large font mode:** about 15% larger font size across the whole app, for older or visually impaired users. A display option, therefore OFF by default.

Good to know

The toggles travel to the Follower as part of the shared diary package: if the Owner switches off, for example, the predicted low alert, it will not sound on the Follower’s phone either. Display settings (language, colours, large font), however, may differ per device.

Emergency (SOS) card

The data of the help page that opens when the red SOS button in the menu is tapped. Everything entered here is stored only on your device.

- **Full name and own phone number:** shown on the SOS page AND in the header of the medical report.
- **Home address:** for helpers and paramedics.
- **Contacts to notify:** several people can be added (name, relation, phone number); they get one-tap call buttons on the SOS page.
- **“How to communicate with me” note:** freely editable text for unknown helpers (e.g. “I have diabetes. If I act confused, give me a sugary drink...”).

Blood glucose limits card

The limits behind alerts, colour feedback and statistics (TIR) — enter the values agreed with your doctor!

-  **Low limit:** below this a value counts as low (alert, amber colour, hypo counter). Default: 3.9 mmol/l.
- **High limit:** above this a value counts as high (red colour). Default: 10.0 mmol/l.
- **Target glucose:** the target of the correction calculation of the bolus calculator. Default: 6.0 mmol/l.

Insulin settings card

Every parameter of the bolus calculator and the IOB calculation.

- **Rapid (bolus) insulin:** dropdown with the rapid insulins available in Hungary (Humalog, NovoRapid, Fiasp, Apidra, Lyumjev...); a custom name can also be typed. This name appears on the forms and in the report.
- **Basal insulin:** likewise from a dropdown (Lantus, Abasaglar, Toujeo, Levemir, Tresiba...), custom name possible.
- **Insulin action time (DIA):** adjustable between 2 and 8 hours, default 4 hours. This is the basis of the insulin-on-board (IOB) calculation — Humalog is typically ~4 hours.
- **Basal injection time:** default 22:00 — the basal insulin reminder aligns to this.
- **Remind this much earlier:** how long before the injection time the purple reminder banner should appear (default: 60 minutes).
- **Reminder on the Overview (toggle):** switches the basal reminder on/off entirely. The reminder NEVER appears for the Follower.
- **Insulin sensitivity (ISF):** 1 unit of rapid insulin lowers glucose by this much. Default: 2.5 mmol/l.
- **Carb/insulin ratio (ICR) by time of day:** separate values for Morning / Noon / Evening — how many grams of carbs need 1 unit of rapid insulin. Default: 10 g/U in all three periods.

Important medical warning

The ICR, the insulin sensitivity (ISF), the target glucose and the insulin action time (DIA) must only be used with values agreed with your treating physician! The bolus calculator computes from these — a wrong value gives a wrong suggestion. Decide about every dose according to your doctor's guidance!

Time-of-day boundaries card

Two independent hour schedules can be set.

- **Meal default boundaries:** five times decide which meal type the app offers for a new entry (until Breakfast, until Morning snack, until Lunch, until Afternoon snack, until Dinner — afterwards Late snack). Defaults: 9 / 11 / 14 / 17 / 21 h.
- **Bolus-ICR boundaries:** two times decide whether the calculator uses the Morning / Noon / Evening ICR. Defaults: Morning until 10 h, Noon until 16 h, Evening afterwards.

Physical activity levels card

The names of the 1–5 scale can be rewritten (defaults: Very light, Light, Moderate, Strenuous, Very strenuous). Your own names appear on the entry form, on the cards and in the report.

Saving and sync

The “Save settings” button stores every change. With Google Drive sync the settings are synchronised between your devices, so the phone and the computer behave the same. In follower mode the Follower may use their own display settings on their own device — this does not affect the Owner’s data.

New in v20.1: when the Activity/heat profile block appears, the temperature is now looked up automatically by default (can be turned off in Settings); the “Activity (choose from earlier ones)” list now appears collapsed by default.

New in v20.1–v20.2: the food row fields switched to Time | Mode (g/ml/pcs) | Amount | Carbs/100g (or carbs/100ml) | Delete; the Mode can be overridden with the dropdown, and for piece/serving-based foods the Amount field is the piece count (behaving just like the old $\times 1/\times 2/\times 3$ multiplier button).

4.8 New entry

Reached with the plus button in the header. The date and time default to the current moment but can be changed freely, so a forgotten morning reading can still be logged in the afternoon with its real time. The type can be Meal, Check (measurement only), Basal (Lantus) or Other activity (e.g. a walk). For meals the application pre-selects the meal type by time of day — based on the boundaries configured in Settings, for example Lunch at noon. Foods are added from the quick picker with one tap and the carbs add up. When a blood glucose value is entered, the application calculates an insulin suggestion (details in chapter 6), which can be accepted with the Apply suggestion button, or a different dose can be entered manually.

The screenshot shows the 'New entry' form in the HBC Diabetes Diary app. The form is titled '+ New entry' and includes the following fields and buttons:

- Time** (can be changed later): 08/17/2026, 04:34 PM
- Type**: Meal
- Meal type**: Choose...
- Blood glucose (mmol/L)**: 9.0. A 'BT' button is next to the field.
- Weight (kg)**: 72.5
- Foods (quick picker)**: Carbohydrates (g). A 'Dose' button is next to the field.
- Fatty, protein-rich meal**: checkbox (unchecked)
- Humalog (U)**: 9.0
- Notes**: Notes...
- Private entry (completely hidden from the followers)**: checkbox (unchecked)
- Save** button (large blue button at the bottom)
- Close** button (red X button at the bottom right)

4.8 New entry — entry form with bolus calculator (illustrated with synthetic sample data).

Separate injection time (v12.3): if the insulin was not injected at the entry's time — for example the bolus was delayed until after the meal because of low blood glucose — the “🕒 Injection time differs?” button under the dose field lets you set a separate time for the rapid and the basal insulin. The ✖ button resets it to the entry time. IOB and every insulin-related statistic, chart and report follow the actual injection time.

Type-specific fields (v14): the fields adapt to the selected type. On Check entries the Extra carbs block (with the full food quick picker) sits before the rapid-insulin correction field — for recording carbs taken at a low reading. On Basal entries only the basal dose (and its injection time) can be entered. On Other activity entries — and on Check entries too — an activity quick picker (from earlier names), a freely typed new name, a from-to duration adjustable in 5-minute steps up to 4 hours and a 1–5 Physical activity indicator are available. Since the activity is typically logged afterwards, once it is finished, the duration slider counts the start backwards from the entry (logging) time (v18.6). The “Set exact start/end” button lets you fine-tune the Start and End clocks independently — e.g. if the entry was logged later than the activity actually ended; the Overview and the entry detail view then also show the start–end time range. On Other activity entries — and only there — the “Private entry” switch can be turned on; such an entry is completely invisible to the follower. Exactly the same fields appear when editing existing entries.

Typo protection (v9.0): if the entered blood glucose value falls outside the plausible range — below 2.0 mmol/l or above 25.0 mmol/l — the application asks before saving whether the value is really correct. This way an accidental 65 instead of 6.5 does not distort the statistics and the report. Answering Yes still saves the unusual value, because rare but real extreme readings do occur.

New in v18: the BT button reads the last measurement from a Bluetooth glucose meter (Android + Chrome only; not supported on iPhone, where manual input remains); when the “Fatty, protein-rich meal” toggle is ticked, the app warns about a prolonged glucose rise and recommends a control measurement in 2–3 hours; the food search also looks in the built-in Hungarian carb table — with category buttons or by typing at least 2 letters.

New in v18 — per-item carb times: a clock field appears in the rows of the foods added to an entry, showing the entry time by default. If you ate an item later (or earlier), set the actual time — the modified item is marked with a green frame, after saving it appears with a mark in the entry details, it is shown as a separate bar at its own time on the carb chart, and the post-meal measurement reminder counts from the time of the last carb item. It can be edited the same way afterwards (pencil icon). The “Other/Extra carbs” counts with the entry time.

4.9 Quick reference: buttons, icons, indicators

The table summarises the recurring elements of the screens to support confident use even in the first days. In v9.0 the icons follow a consistent drawn style.

Element	Where it is found	What it does
Plus button (New)	Header, top right	Records a new entry from any tab.
Bottom navigation bar	On phones, at the bottom of the screen	Switches between the seven tabs; the active tab is coloured.
Pencil icon	On entry cards	Edits the entry afterwards, including its time.
Bin icon	On entry cards	Deletes, after a confirmation question.
Green, amber, red colours	On cards and charts	In-range, low, and high blood glucose values.
Pulsing IOB label	Overview, insulin tile	Insulin is currently active in the body; the amount is shown in units.
Purple bar under the header	In follower mode	The follower's view: the time of the last sync and the Sync now button.
Apply suggestion button	New entry, purple box	Moves the calculated insulin dose into the field, after confirmation.
Create report button	Bottom of the Statistics tab	Report panel for the given from-to period: send it by e-mail to your doctor (with a saved address list), or PDF / print.
Sync now button	Sync tab and follower bar	Immediate manual upload or refresh through Drive.

5. Diabetes, insulin and nutrition — what's worth knowing

This chapter does not replace the personalized guidance of your treating physician or diabetes nurse educator — it gives general, informational background knowledge for people using the HBC Diabetes Diary. If you're unsure about anything here, always check with your care team.

5.1 Type 1 diabetes in brief

In type 1 diabetes, the immune system mistakenly destroys the insulin-producing beta cells of the pancreas. As a result, the body produces essentially no insulin of its own, so people with type 1 diabetes need externally supplied insulin for the rest of their lives — this is not a lifestyle-related condition, and it cannot be prevented through diet or exercise.

This is fundamentally different from type 2 diabetes, where the body usually still produces insulin, but its effect is reduced (insulin resistance), and it can often be managed with diet, exercise, or oral medication. Type 1 diabetes can develop at any age, most often in childhood or young adulthood. Without insulin, blood glucose rises uncontrollably, and a severe, life-threatening condition (diabetic ketoacidosis) can develop within a short time — which is why insulin replacement is not optional, but life-sustaining, ongoing treatment.

5.2 How insulin works — the basal-bolus principle

The most commonly used insulin regimen today follows the so-called basal-bolus principle, which tries to mimic the pancreas's natural function.

- **Basal insulin (e.g. Lantus, Toujeo, Tresiba, Levemir):** absorbed slowly and evenly, providing the body's baseline insulin need around the clock, independent of meals — this is what the diary's "Lantus" entry type records.
- **Bolus (rapid-acting) insulin (e.g. Humalog, NovoRapid, Fiasp, Apidra):** absorbed quickly and acts for a short time; it covers the carbohydrate content of meals (meal bolus), and when needed, also corrects a high glucose reading (correction bolus).

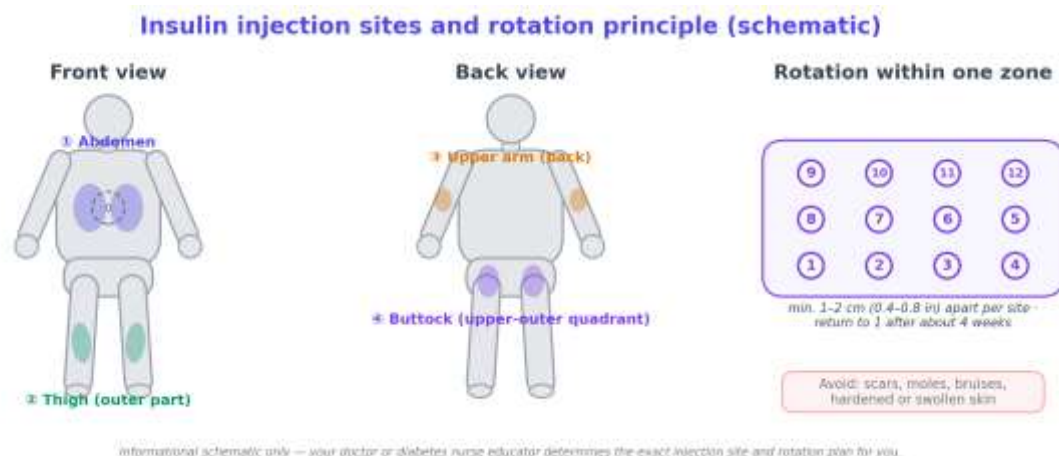
The diary's bolus calculator combines these two needs — the meal-related dose calculated from your configured carb-to-insulin ratio (ICR), and the correction dose calculated from your insulin sensitivity factor (ISF) — into a single suggestion, which is always informational only; the final decision is yours, following the rules you've agreed on with your physician.

5.3 Insulin injection technique, step by step

Correct injection technique is one of the basic requirements for effective, predictable insulin action. The steps below provide general information; your physician or diabetes nurse educator will always demonstrate and check the exact technique with you in person.

- **Wash your hands**, then keep the chosen skin area clean and dry (alcohol disinfection is only needed if the skin is visibly soiled).
- **Check the insulin:** expiry date, and whether rapid-acting insulin is clear while basal insulin (depending on type) is evenly cloudy; insulin that has been frozen or chilled to ice-cold should not be used.
- **Choose the injection site:** for the abdomen, avoid the roughly 5 cm (2 inch) band around the navel; on any body area, never inject into scarred, mole-covered, bruised, hardened, or swollen skin.
- **Change the needle before every injection**, then prime the pen (dial and expel 2 units until a drop appears at the needle tip).
- **Pinch a skin fold if needed** (this can be essential for thinner body types, children, or shorter needles) — using two fingers, without picking up muscle.
- **Insert the needle into the skin** (at an angle of 45–90 degrees, depending on needle length and skin fold), then press the dose in steadily, without rushing.
- **Keep the needle in the skin for about 10 seconds** after delivering the dose, before withdrawing it — this reduces the chance of leakage.
- **Remove the needle safely and dispose of it in a sharps container;** gently massaging the injection site afterward is NOT recommended.

Rotation: within the same body area, each injection should be at least 1–2 cm away from the last one, and you should return to a given spot only after about 4 weeks. Repeated injections into the same spot can cause a subcutaneous fat change (lipohypertrophy), which makes insulin absorption worse and less predictable. The schematic below illustrates the most commonly used body areas and one possible rotation pattern:



Original schematic illustration — your care team always determines the exact injection site and rotation plan.

Absorption speed by body area (informational order, fastest to slowest): abdomen → upper arm → thigh → buttock. This is why many people prefer the abdomen for rapid-acting insulin, while the thigh or buttock is often chosen for basal insulin, for slower, more even absorption — but the exact recommendation can differ from person to person, so discuss this with your care team as well.

5.4 Hypoglycemia and hyperglycemia — recognition and response

Low blood glucose (hypoglycemia): generally a reading below 3.9 mmol/l (70 mg/dl). Typical symptoms: shakiness, sweating, hunger, rapid heartbeat, paleness, irritability, difficulty concentrating, confusion; in severe cases, loss of consciousness can occur.

What to do (the “15-15 rule”): check your blood glucose, then eat about 15 grams of fast-acting carbohydrate (e.g. 3 glucose tablets, 150 ml/5 oz of fruit juice, or a non-diet soft drink) — NOT fatty or protein-rich food, since that slows absorption. Recheck after 15 minutes; if the level is still low, repeat. If the low occurs at least 1 hour before a meal or planned exercise, it's also worth having a small, complex carbohydrate snack once you've recovered.

If someone loses consciousness

never give anything by mouth — glucagon (injection or nasal spray) and/or emergency medical help is needed. The HBC diary's SOS feature is designed to help prepare for this (see Chapter 14).

High blood glucose (hyperglycemia): the exact threshold varies by person, but readings above 10 mmol/l (180 mg/dl) are often considered elevated, and above 13.9 mmol/l (250 mg/dl) significantly elevated. Persistently high readings — especially together with nausea, abdominal pain, rapid breathing, or a distinctly fruity-smelling breath — call for a ketone check and medical consultation, since this can be a sign of diabetic ketoacidosis, which needs urgent care. For an occasional, isolated high reading, follow the correction rule you've agreed with your physician; the diary's bolus calculator helps here in an informational way, but does not replace individual medical guidance.

5.5 Carbohydrate counting basics

Carbohydrate counting (carb counting) is the basis for determining the meal insulin dose: the more carbohydrate a meal contains, the greater the rise in blood glucose, and the more rapid-acting insulin is needed to cover it.

In the HBC diary's food database, every food has a carbohydrate value given per 100 grams (or 100 ml); when you enter the amount actually eaten, the app automatically calculates the item's actual carb content. A new food is easy to add using the “carbohydrate” line of the nutrition label on the package (not the “of which sugars” line — that's only a subset); for unfamiliar, composite meals (for example, at a restaurant) the estimate is naturally less precise, and in that case a reasonable estimate plus learning afterward (what your glucose did given that particular estimate) helps refine future decisions.

The carb-to-insulin ratio (ICR) — how many grams of carbohydrate 1 unit of rapid-acting insulin covers — varies from person to person, and often by time of day as well; it is always set by your physician, and can be saved in the diary's Settings so the bolus calculator can use it.

5.6 The effect of exercise on blood glucose

Regular physical activity has many benefits in type 1 diabetes as well (cardiovascular health, improved insulin sensitivity, wellbeing), but its effect on blood glucose is less predictable than that of food or insulin. Aerobic exercise (e.g. running, cycling, swimming) typically lowers blood glucose — sometimes for hours after the activity ends (delayed hypoglycemia risk, typically that same night or the following night). Intense, short, high-effort exercise (e.g. weightlifting, sprinting), by contrast, can temporarily raise blood glucose due to stress-hormone release.

- **Before exercise**, check your blood glucose, and if it's low or at the lower edge of your target range, eat some carbohydrate first.
- **For longer exercise (over 60 minutes)**, check hourly, and top up carbohydrate as needed.
- **If the exercise is planned**, your insulin dose (basal and/or bolus) can be reduced beforehand, as agreed with your physician.
- **Always carry** fast-acting carbohydrate with you during exercise as well.

The HBC diary's Activity/heat profiles (Settings) help you assign your own, experience-based percentage correction to the bolus calculator for activities you do regularly — the Statistics page's retrospective analysis helps fine-tune these percentages over time.

5.7 “Sick day” rules (fever, vomiting)

During a feverish illness, infection, or an episode of vomiting/diarrhea, blood glucose becomes harder to predict, and — contrary to what many people assume, thinking they need less insulin because they're eating less — in type 1 diabetes the body's stress response usually means MORE insulin is needed, not less, even with reduced appetite.

Never stop basal insulin on your own on sick days

even if you're eating very little — stopping it can lead to rapid, serious deterioration.

- **Check blood glucose more often** (as often as every 2–4 hours); when feverish, vomiting, or having diarrhea, always consider checking ketones as well, if you're able to.
- **If you can't eat solid food**, replace carbohydrate with liquids (e.g. sweetened tea, fruit juice) matching your usual meal amounts.
- **Drink plenty of fluids** (water, sugar-free liquids) to avoid dehydration.
- **Warning signs needing immediate medical care:** persistent vomiting, high blood glucose with ketones, or if the person becomes increasingly drowsy or confused.

It's worth discussing and writing down a personalized “sick day” plan with your physician in advance, while healthy (when and how much to increase correction insulin, when to seek medical care) — this can be kept on paper or in the HBC diary's notes field, so it's quickly available if you do get sick.

6. Bolus calculation in detail

This chapter walks through, step by step and with worked numbers, exactly how the diary's bolus suggestion is calculated — so that you don't just accept the number, but understand what's behind it. Like the previous chapter, this is general, informational background knowledge, not a substitute for the personalized guidance of your treating physician or diabetes nurse educator — always set your own ICR, ISF and effect-duration values together with them, on the Settings (4.7) tab.

6.1 The formula in one place

Every bolus suggestion in the diary follows the same internationally common, three-part formula:

Suggested dose = Carb-coverage dose + Correction dose – Active insulin (IOB)

- **Carb-coverage dose:** the insulin needed to cover the carbohydrate eaten (6.2).
- **Correction dose:** insulin that adjusts for the gap between the measured value and the target — positive if the reading is above target, negative if below (6.3).
- **Active insulin (IOB):** the estimated amount still working from an earlier dose, subtracted to avoid stacking doses on top of each other (6.4).

The result is rounded to the nearest 0.5 unit and can never be negative — in the worst case the suggestion is 0. The sections below break the formula down term by term, then walk through one complete example.

6.2 Carb-coverage dose (ICR)

ICR (insulin-to-carb ratio) expresses how many grams of carbohydrate 1 unit of rapid-acting insulin covers. The higher the ICR, the less insulin is needed for the same meal — so this value should only ever be changed together with your physician. The diary allows a separate ICR for each time of day (Settings, tab 4.7), since many people are more insulin-sensitive in the morning than at midday or in the evening.

Carb-coverage dose = carbohydrate eaten (grams) ÷ ICR (grams/unit)

- **Example:** 50 grams of carbohydrate, with a breakfast ICR of 8 g/U: $50 \div 8 = 6.25$ units.

6.3 Correction dose (sensitivity factor, ISF)

ISF (insulin sensitivity factor) shows how many mmol/l 1 unit of rapid-acting insulin lowers blood glucose by. The correction dose is calculated from the gap between the measured value and the configured target: if the reading is above target, it adds a positive amount of insulin to the suggestion; if below, it subtracts (in practice this means less insulin is suggested than the carbs alone would call for).

Correction dose = (measured value – target value) ÷ ISF

- **Example:** a measured value of 11.0 mmol/l, a target of 6.5 mmol/l, and an ISF of 2.0: $(11.0 - 6.5) \div 2.0 = 2.25$ units.

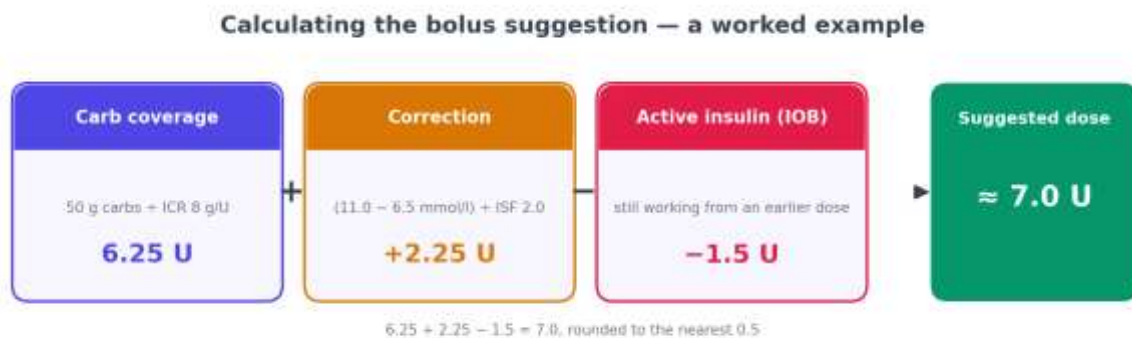
6.4 The role of active insulin (IOB)

Rapid-acting insulin doesn't act all at once — it works along an effect curve over several hours (the length of this curve is set in Settings, based on medical advice). Active insulin (IOB, “insulin on board”) is the estimated amount still not yet broken down from an earlier dose — the diary automatically subtracts this from a new suggestion, to help avoid dangerously stacking doses on top of each other. This is an informational estimate, not a precise physiological measurement: if your symptoms or a fresh measured value don't match what the number suggests, always go by the measurement and the symptoms, not solely by the displayed IOB.

- **Example:** if an estimated 1.5 units are still active from an earlier dose, the diary factors this in with a minus sign in the next suggestion.

6.5 One complete worked example

The figure below breaks a single dosing decision down into its three terms, then sums the result. The numbers are illustrative — not real user data.



Illustrative example only — always set your own ICR / ISF / effect-duration values in Settings together with your physician. The final decision is always yours.

6.5 Calculating the bolus suggestion — a worked example (illustration, original artwork).

In this example, 50 grams of carbohydrate were eaten (ICR 8 g/U), the reading was 11.0 mmol/l (target 6.5 mmol/l, ISF 2.0), and 1.5 units were still estimated active from an earlier dose. The calculation: 6.25 (carb coverage) + 2.25 (correction) - 1.5 (IOB) = 7.0 units — this lands exactly on a half unit, so no further rounding is needed and 7.0 is the final suggestion. The diary's Apply suggestion button carries this straight into the insulin field with a single tap, but the number can always be overridden by hand.

6.6 When and why the suggestion can be overridden

The calculated number is always just a suggestion — the final decision is always yours, based on principles agreed with your treating physician. A few typical situations where users deviate from the suggested number, based on their own experience or their care team's advice:

- **Illness or fever:** insulin needs often rise at these times, and separate rules apply — see section 5.7.
- **Unusual or planned physical activity:** the effect of exercise on blood glucose can change hour to hour — see section 5.6.
- **An uncertain carb estimate:** for example at a restaurant or with unpackaged food, where the gram count is only an approximation.
- **Symptoms that don't match the number:** if weakness, shakiness, or another hypoglycemia symptom appears, always go by the symptom and a fresh measurement, not by the displayed suggestion.
- **Device or measurement uncertainty:** for an unusually high or low reading, it's worth repeating the measurement before applying the suggestion.

In case of severe hypoglycemia

if someone becomes confused or loses consciousness, act immediately — never give anything by mouth in this situation. The diary's SOS feature (Chapter 14) is designed to help prepare for exactly this kind of situation.

7. A typical day with the application — a practical example

This chapter walks through an ordinary weekday from morning to evening through the eyes of an example user, to show where the diary fits into everyday life. The numeric values are illustrations; personal parameters must always be agreed with the treating physician.

7:00 am — waking measurement and breakfast

The user measures their blood glucose: 6.2 mmol/l. They press the plus button. The type is Meal and the meal type already says Breakfast. They enter 6.2, then pick from the quick picker: 2 slices of wholemeal toast (24 g carbs) and a medium apple (15 g carbs), 39 grams in total. The application calculates with the morning ICR (10 g/U in this example): 3.9 units; the deviation from target is small so the correction is negligible, there is no active insulin, and the suggestion is 4.0 units of rapid insulin. The user accepts the suggestion, injects the insulin, then Save. The Overview shows: 1/4 entries today, the streak continues.

12:30 pm — lunch with higher glucose

Reading: 9.0 mmol/l; lunch: chicken with rice, about 60 grams of carbs. The calculation has three parts. Carb part: $60 / 10 = 6.0$ units. Correction: $(9.0 \text{ measured} - 6.0 \text{ target}) / 2.5 \text{ sensitivity} = 1.2$ units. Active insulin: nothing remains from the morning dose, so 0. The suggestion: $6.0 + 1.2 = 7.2$, rounded to half units: 7.0 units. The application also displays the details, so the follower can see afterwards exactly how the dose was composed.

4:00 pm — treating hypoglycaemia

The user feels weak and measures: 3.4 mmol/l. They act first: 15 grams of glucose, 15 minutes of waiting, re-check: 4.6 mmol/l. Then they record the 3.4 value with the Check type. On saving, the application confirms the protocol in a pop-up (15 grams of fast carbs, re-check in 15 minutes), the value appears as an amber dot on the chart, and the follower's phone receives an alert if it has been enabled.

9:30 pm — basal insulin and closing the day

The Overview signals in a purple bar: no basal insulin logged today. The user injects their usual basal dose and records it as a Basal-type entry. The daily circle turns 4/4, the weekly TIR tile shows 81%, and the day ends with a green confirmation.

Weekend example: exercise and a restaurant

On Saturday morning the user cycles for an hour and a half. Before setting off they measure 7.8 mmol/l, and because exercise lowers blood glucose, they eat 20 grams of carbohydrate without insulin according to the protocol agreed with the physician. They record this as an Other activity entry with the note: before 90 min cycling. In the evening they eat out. The exact carb content is an estimate in such cases: the user takes 70 grams as the basis of the bolus for the pasta dish, based on experience. After saving they write in the note: restaurant estimate, carbonara. Two hours later they log a check reading, 9.6 mmol/l. When the application calculates the next suggestion, it automatically subtracts the insulin still on board (IOB), reducing the danger of stacked doses. Days annotated like this are especially valuable at the clinic: the treating physician can see exactly which situations cause the swings.

Before the quarterly check-up: a report in two taps

On the evening before the appointment, the user opens the Medical report card on the Statistics tab, sets the past 90 days, presses Create report and saves the summary as a PDF. The next day the physician sees the TIR, the estimated HbA1c and GMI, the variability measures, the daily pattern and the full log on a single page — so the consultation can focus on decisions.

8. The phases of installation

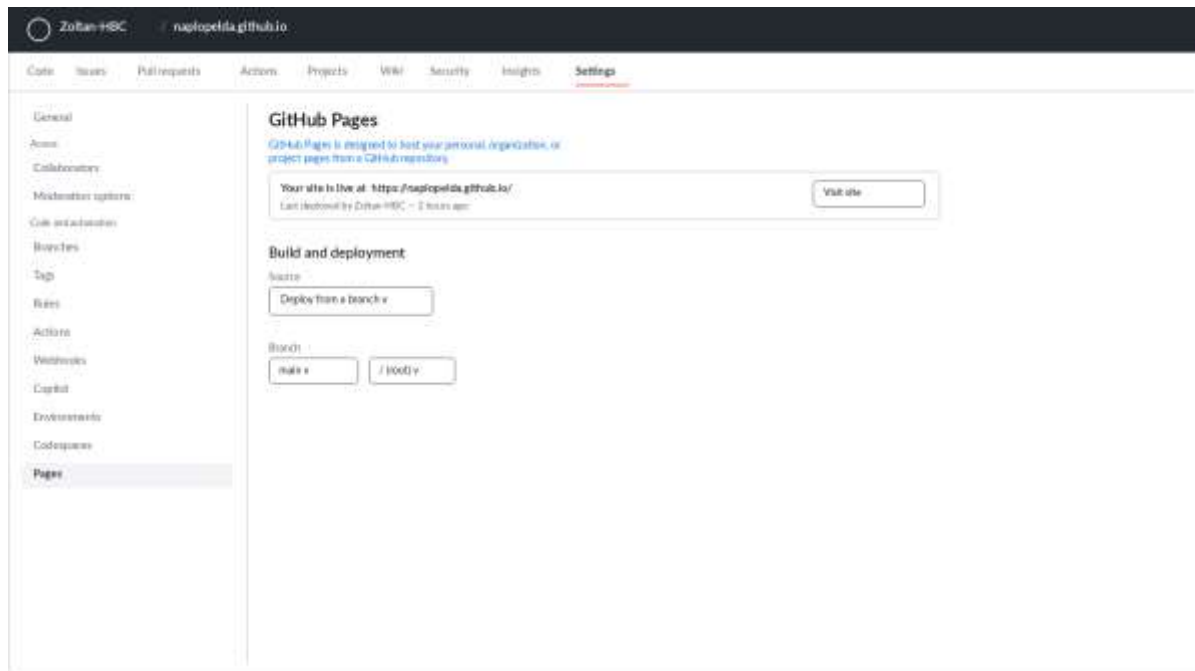
Installation has two big steps: first the application must be published once on the internet (it is sufficient for a single person to do this), then everyone installs it on their own device from a browser. The reason for publishing: for security reasons, installability and Drive sync only work on an encrypted, so-called HTTPS address — not when the file is opened directly from the computer.

8.1 First phase: publishing on GitHub Pages

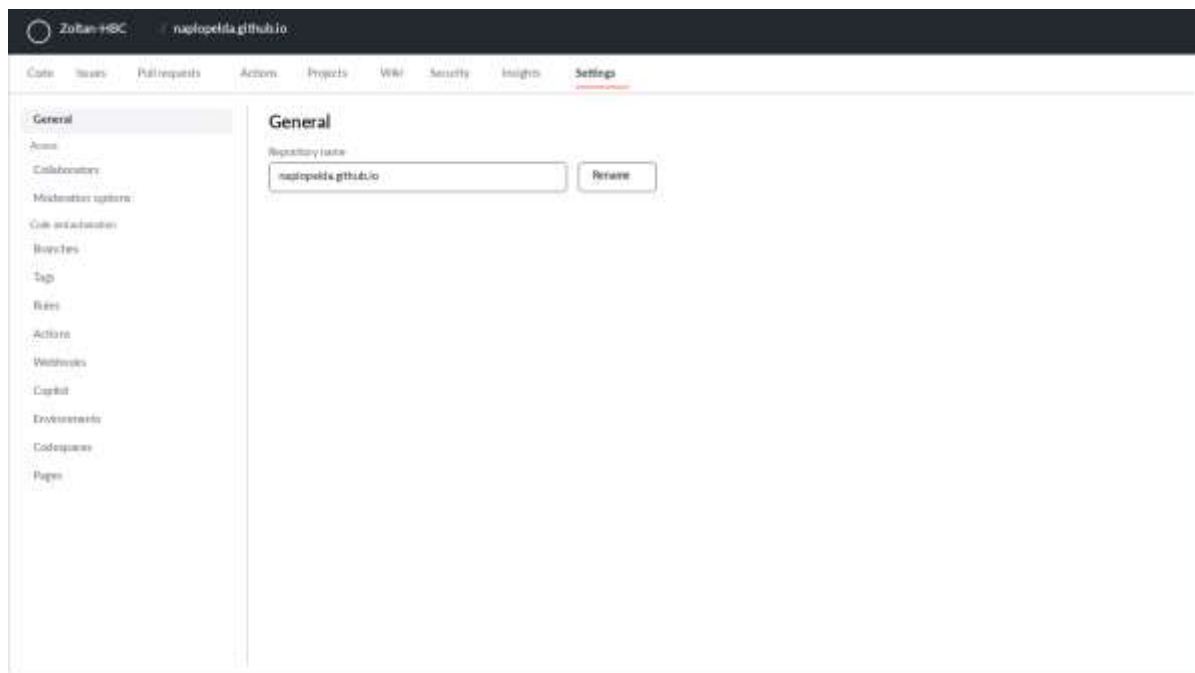
GitHub is the world's largest free code host, and its Pages service provides a free web address. No bank card is needed and the whole process takes about 15 minutes. Based on practical experience, the steps below highlight the two most common pitfalls: the exact repository name and the correct way of uploading the files.

- Open github.com and sign up (an e-mail address and a password are required).
- In the top right corner click the plus sign, then New repository.
- **The repository name must be exactly the full web address: YOURUSERNAME.github.io — including the .github.io ending! As an example, using a fictional username naplopelda: correctly, the repository name is naplopelda.github.io, and the application will be available at the full address https://naplopelda.github.io. Incorrect examples: just naplopelda (without the .github.io ending), or any other name that does not exactly match the username (e.g. naplopelda-site) — with these the Pages service will not start at the main address.**
- Set it to Public, then click Create repository.
- **Verification: in the repository's Settings → General section the name (in the example above: naplopelda.github.io), and on the Settings → Pages screen the web address, must both appear in the full https://naplopelda.github.io form — together with the .github.io part. If the ending is missing in either place, the name given in step 2 is wrong and the repository must be renamed (Settings → General → Repository name).**
- Click the “uploading an existing file” link.
- **When uploading, drag in ALL files and subfolders that are INSIDE the HBC_App_v12 folder — but NOT the folder itself! Correctly: open the HBC_App_v12 folder on your computer, select everything in it (Ctrl+A) and drag this selection into the browser window: the index.html, manifest.json and sw.js files and the css, js, lib, fonts, icons and store folders. Incorrectly: if you drag in the HBC_App_v12 folder as a whole, the application ends up under the .../HBC_App_v12/ sub-address and will not start at the main https://YOURUSERNAME.github.io address.**

k) GitHub — Settings → Pages: the “Your site is live at...” confirmation



l) GitHub — Settings → General: the exact repository name can be checked here



- Click Commit changes and wait 2–3 minutes.
- Open the full address <https://YOURUSERNAME.github.io> in a browser. If the application loads, publishing is complete.

Good to know

The public repository contains only the application’s program files, not the diary data. The data is always stored on the individual phones and in the user’s own Google Drive.

8.2 Installing on an Android phone

- Open <https://YOURNAME.github.io> in the Chrome browser.
- An Install offer pops up at the bottom of the screen. If not, tap the three dots in the top right, then Install app.
- Tap Install. The icon (the blue-grey ribbon logo) is placed on the home screen.
- Open it from the icon: the application runs full-screen and from then on works without internet as well.

8.3 Installing on an iPhone

- Open <https://YOURNAME.github.io> in the Safari browser.
- Tap the Share button at the bottom (a square with an upward arrow).
- Scroll down, choose Add to Home Screen, then tap Add.
- The icon appears on the home screen; started from there it runs full-screen, even without internet.

8.4 Installing on a Windows or Mac computer

- Open <https://YOURNAME.github.io> in the Chrome or Edge browser.
- A small install icon appears at the right edge of the address bar (a monitor with a downward arrow). Click it.
- Click Install. The application runs in its own window and is added to the Start menu or the Dock.

8.5 Taking over the data of an earlier (v6, v7 or v8) diary

There are two cases. If the new application is opened in the same browser in which the v6, v7 or v8 diary was used so far, the data appears by itself — nothing needs to be done. When starting on another device or browser: in the old diary open the Sync tab, press JSON Export, send the file to yourself (for example by e-mail), then on the Sync tab of the new application load it with the JSON/CSV Import button. The import is intelligent: it does not duplicate entries that already exist.

8.6 First settings, together with the treating physician

After installation open the Settings tab and complete the table below. The values in the left column are approved by the treating physician, because the insulin suggestions are calculated from them.

Setting	What it means	Example value (to agree with the physician)
Low blood glucose limit	The hypo warning sounds below this	3.9 mmol/l
High blood glucose limit	Values above this count as high	10.0 mmol/l
Target glucose	The target of the correction calculation	6.0 mmol/l
Sensitivity	The glucose-lowering effect of 1 unit of insulin	2.5 mmol/l
ICR morning Insulin-to-Carbohydrate Ratio (ICR)	How many grams of carbs 1 unit covers in the morning	10 g
ICR midday Insulin-to-Carbohydrate Ratio (ICR)	How many grams of carbs 1 unit covers at midday	12 g
ICR evening Insulin-to-Carbohydrate Ratio (ICR)	How many grams of carbs 1 unit covers in the evening	10 g
Time-of-day boundaries	The hour limits of meal types and ICR periods	Breakfast until 9, Lunch until 14, Dinner until 21; ICR: morning until 10, midday until 16
Insulin action time	How long injected rapid insulin counts as active	4 hours
Insulin names	The names of the preparations used	e.g. Humalog and Lantus
Custom background color (v10.0)	A background color choosable from a pop-up palette next to Settings → Appearance	e.g. #EEF2FF (clear anytime with Reset)

8.7 Updates in the future

When a new version of the application is ready, updating is simple: the changed files are uploaded to the GitHub repository the same way as in chapter 8.1. The copies installed on phones and computers notice the new version by themselves: they download it the next time they are opened with an internet connection, and the version after that restart is already the new one. Diary data remains untouched during the update, because it lives not in the program files but in the device's own storage. Example: files uploaded on Tuesday evening are downloaded by the follower's phone when opened on Wednesday morning, and the second opening that morning already runs the new version.

“App update available” notice: on an installed application the browser (Chrome/Edge) may show its own built-in notice at the top of the screen when it detects a new version. This is not a message from the application and not an error: the diary is typically already running the new version when the notice appears, and clicking it merely refreshes the browser's own records. The notice can safely be clicked away; it cannot be switched off because it is a browser feature, but it causes no harm and does not touch the data.

9. Google Drive sync: the follower sees the values live

9.1 How it works

Sync is built on two roles. The owner's application automatically uploads the diary as a single file to a chosen folder of their own Google Drive after every save. The follower's — a relative's or the treating physician's — application connects to this shared file: it downloads fresh data on opening and at a configurable interval (5 minutes by default), and shows it on the same screens as the owner's — just read-only. Alerts can be requested too: when a fresh low or high value arrives, the follower's phone shows a notification; from v14 a notification can optionally be requested for every new entry as well. Alerts arrive while the application is open or in the background; with the application fully closed they do not — that would require a central server, which may be a future extension.

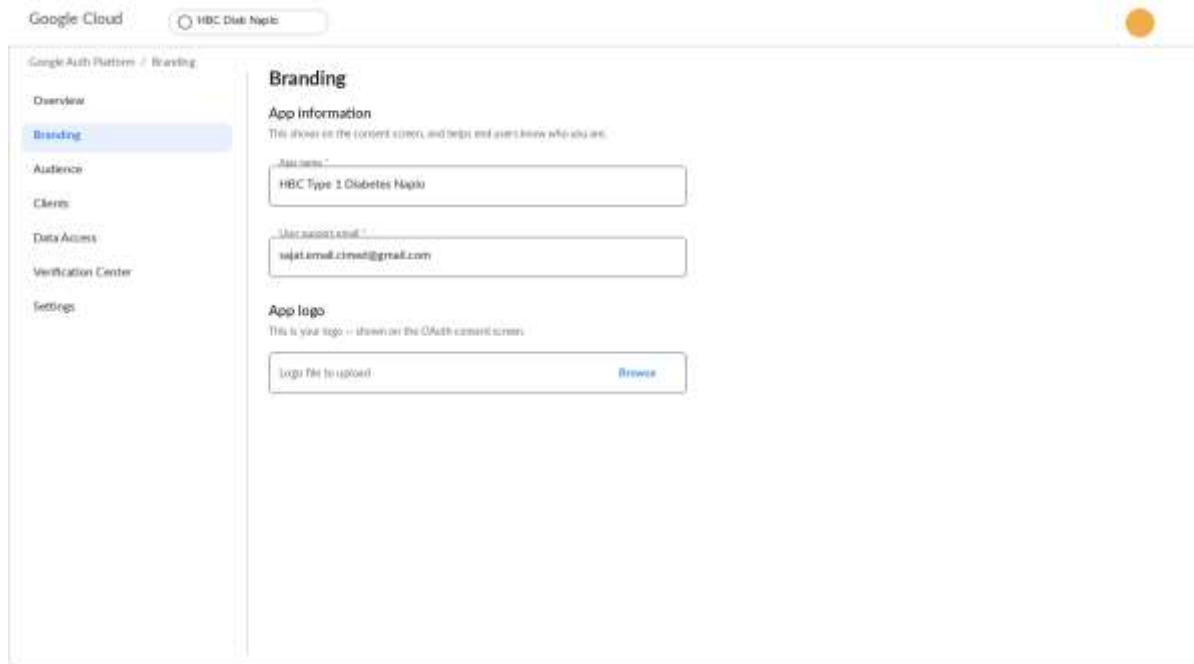
The application uses Google's strictly limited permission called `drive.file`: it can access only its own diary file and cannot even see the other files stored on the Drive.

9.2 One-time Google setup (about 20 minutes, free)

Google requires every such application to have a so-called Client ID. It must be created once, without a bank card.

- Open console.cloud.google.com and sign in with your Google account.
- At the top, in the project selector, click New project, name it HBC Naplo, then Create.
- In the left menu: APIs & Services, then Library. Find and enable (Enable) the Google Drive API, then the Google Picker API the same way.
- On first entry the Google Auth Platform setup wizard appears (Overview tab, Get started button). Enter the app name (e.g. HBC Diabetes Diary) and your email address (User support email); on the next step choose the External audience type, enter your email again as the contact, and accept the terms (Finish, Continue).
- In the left-hand Clients menu click + Create client. Set the type to Web application; any name is fine (e.g. Web client 1). In the Authorized JavaScript origins field enter your own address: <https://YOURNAME.github.io>, then Create.
- Click the created client to reveal the Client ID (a long string ending in `.apps.googleusercontent.com`) — copy it, it will be needed in step 7.3.
- In the left-hand Audience menu scroll to Test users and use + Add users to add your own email address and the follower's email address. In Testing publishing status up to 100 test users can be added, no verification required.

g) Google Auth Platform — Branding (app name, contact email)



Google Cloud HBC Diabetes Diary

Google Auth Platform / Branding

Branding

App information
This shows on the consent screen, and helps end users know who you are.

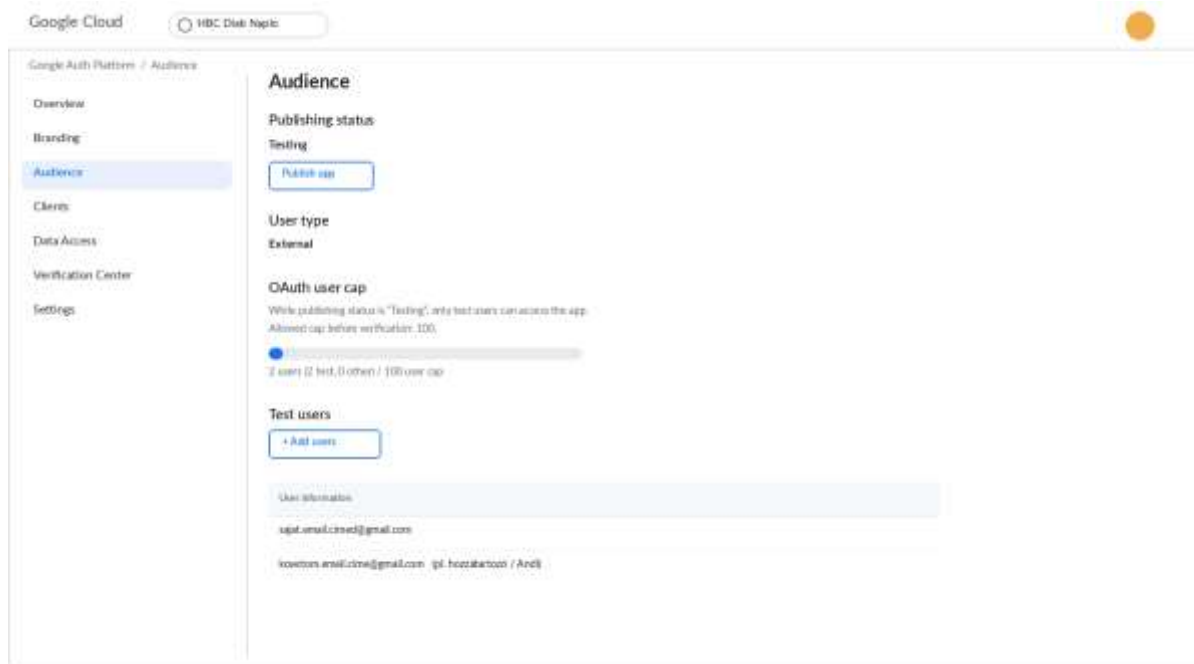
App name
HBC Type 1 Diabetes Diary

User support email
support@hbc.com

App logo
This is your logo - shown on the OAuth consent screen.

Logo file to upload [Browse](#)

h) Google Auth Platform — Audience (Testing status, test users)



Google Cloud HBC Diabetes Diary

Google Auth Platform / Audience

Audience

Publishing status
Testing
[Publish app](#)

User type
External

OAuth user cap
While publishing status is "Testing", only test users can access the app.
Allowed cap before verification: 100

2 user (2 test, 0 other) / 100 user cap

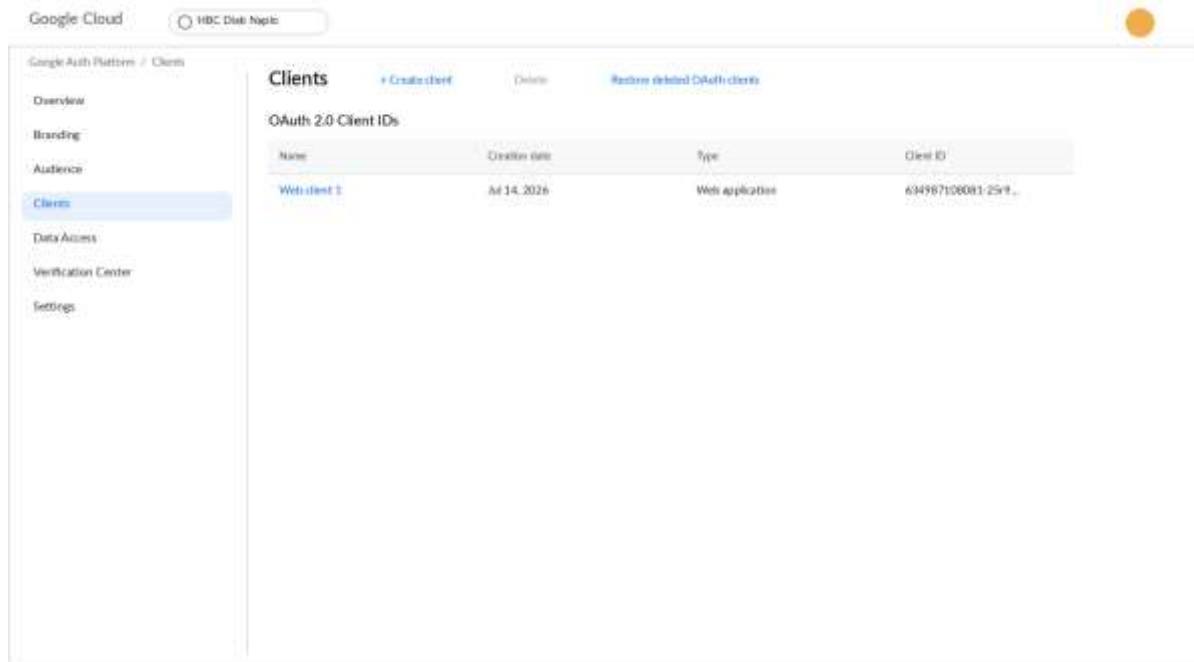
Test users
[+ Add users](#)

User information

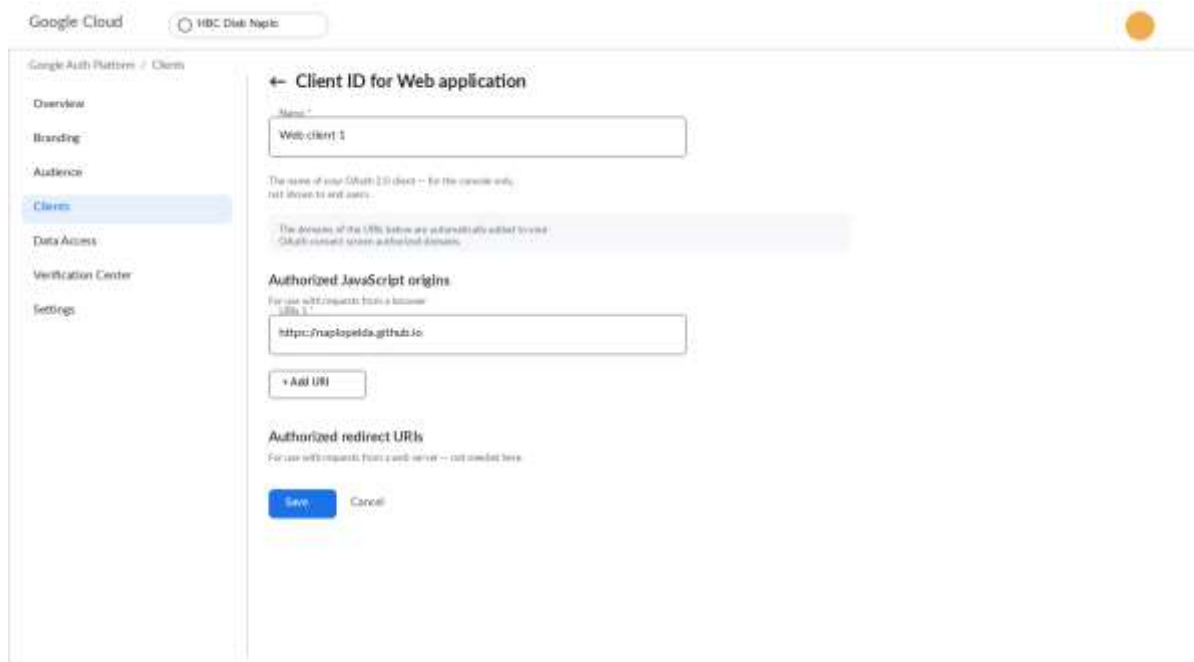
support@hbc.com

support@hbc.com / iOS, Android

i) Google Auth Platform — Clients (the created OAuth client)



j) Client details — Authorized JavaScript origins and the Client ID



9.3 Setting up the owner (owner mode)

- In the HBC Diabetes Diary application itself — NOT in Google Cloud, in the diary app — open the Sync tab, and on the Google Drive sync card choose the Owner mode button.
- Paste the Client ID into field 1.
- Press button 2, Connect with Google account, and allow access. If the Google Cloud project is still in Testing status, Google may show an “unverified app” screen: proceed with Continue, because only the registered test users have access. In production status this screen does not appear.
- Press button 3, Choose folder, and select the diary folder on the Drive (worth creating in advance, for example named HBC Naplo).
- Done: from then on the file on Drive refreshes a few seconds after every save. It can also be triggered manually with Sync now.

9.4 Sharing the folder and setting up the follower (follower mode)

- On drive.google.com right-click the diary folder, choose Share, enter the follower’s e-mail address with the Viewer role, and send it.
- The follower installs the application on their own phone the same way (chapter 8.2 or 8.3).
- On the Sync tab the follower chooses the Follower mode button, pastes the same Client ID, and connects with their own Google account.
- Tapping the Choose shared diary button, they select the HBC_Diabetesz_Naplo.json file from the Shared with me section of the picker.

Important (v12.1): at the Connect with Google account button, Google now asks for the "See your Google Drive files" (read-only) permission. This is needed so that the selected diary file is not "lost" after the Google sign-in expires — even with this permission the app cannot write to or modify anything on Drive. Tip: if Google keeps showing the full consent screen every week, set the project from "Testing" to "In production" (Publish app) on the Google Cloud Console "OAuth consent screen" page — after that the consent is durable.

- In the Alert settings box they enable notifications, switch on the low-value alert, and set the refresh interval (for example 5 minutes). If desired, they also switch on “Notification for every new entry” (v14) — then every new entry triggers a notification, private entries never do.
- Done: the purple bar under the header shows the time of the last sync, and the Sync now button requests an immediate refresh.

An everyday example

The owner records a 3.4 mmol/l value at 16:05. The diary is uploaded to Drive at 16:05. The follower’s phone downloads it at the next refresh, by 16:10 at the latest, and since the value is below the 3.9 limit, it shows an alert: HBC Diary, low value alert, 3.4 mmol/l.

Important (v9.0): the alert fires only for the latest measurement taken on the current day. A back-dated value entered for an earlier day never triggers an alert, so there is no false night-time wake-up because of a past record being filled in. The shared data package also includes the CGM sensor readings, so the alert always reflects the real, current state.

9.5 Backup in practice: the monthly routine

Drive sync gives continuous protection, but it is worth adding a simple monthly routine, because a manual backup is independent even of the Google account. The suggested procedure on the first Sunday of every month: on the Sync tab press JSON Export; the downloaded file name automatically contains the date (for example HBC_v9_backup_2026-08-02.json); copy it to a USB stick or send it to yourself by e-mail. If everything ever needs restoring (new phone, lost device, cleaned browser), the latest such file can be loaded back within minutes with the JSON/CSV Import button, and the import does not duplicate existing entries. The Overview screen warns when the latest backup or sync is older than 7 days, so skipping the routine does not go unnoticed.

10. Loading CGM data (for sensor users)

When using a continuous glucose sensor (for example FreeStyle Libre or Dexcom), the measurement file downloadable from the manufacturer's system can be loaded into the application and appears as a separate curve on the Charts tab. There is currently no live, direct connection to the sensor, because the manufacturers do not open this to third-party applications, but the application is prepared for it, so a later extension can add it.

LibreView example

- Sign in at libreview.com and save the CSV file with the Download glucose data button.
- On the application's Sync tab press the Choose CGM CSV file button and select the downloaded file.
- The confirmation shows the number of imported readings, for example: CGM: 1340 readings imported.
- The CGM sensor curve appears on the Charts tab, with the target-range band.

Dexcom Clarity example

At clarity.dexcom.eu a CSV file can be requested with the Export button; loading works the same way. The application recognises the file's unit: values saved in mg/dl are converted to mmol/l automatically, so the data is stored consistently.

11. Publishing in the Google Play store

The application can also be uploaded to the Play store, so anyone can install it from the store by searching. This is not a requirement for use (installation as in chapter 8 is fully featured); it is rather the path of an official release for a wider audience. The materials needed for the process are ready in the HBC_App_v12/store folder: store texts in Hungarian and English, a privacy statement, and a packaging configuration file.

11.1 Creating a developer account

- Open play.google.com/console/signup and sign in with a Google account.
- Choose the Personal account type and pay the one-time 25 dollar registration fee by bank card.
- Complete the identity verification (a photo of an ID card). Approval usually takes a few days.

11.2 Building the Android package

A packaged Android file (AAB) is uploaded to the Play store; it is produced from the published application by the free tool Bubblewrap. This requires Node.js on the computer (nodejs.org, LTS version).

- In a command prompt run: `npm install -g @bubblewrap/cli`
- Then: `bubblewrap init --manifest=https://YOURNAME.github.io/manifest.json`. For the questions, the values in the store folder's `tw-manifest.json` are authoritative.
- The process creates a signing key: write down and keep its passwords, they will be needed for every future update.
- Run: `bubblewrap build`. The result is the `app-release-bundle.aab` file.
- Bubblewrap prints content named `assetlinks.json`: upload it to the website into the `.well-known` folder. Without it a browser bar would show inside the app.

11.3 Upload and store listing

- In the Play Console click Create app; the name is HBC Diabétesz Napló, the language Hungarian.
- In the Internal testing track create a release and upload the AAB file. This gives an immediately installable link for trying it out before the public release.
- Copy the texts of `store/STORE_ANYAGOK.md` to the store listing (short and long description in Hungarian and English), upload the 512-pixel icon and the screenshots.
- On the data safety form declare: the application does not collect and does not share data with third parties. Provide the web address of the privacy statement.
- In the health declaration state that the application is not a medical device.
- After successful testing create the production release and submit it for review. Assessment typically takes 1–7 days.

Apple App Store note: for iPhones, installation as in chapter 8.3 is the recommended route. An App Store release requires an Apple developer account (99 dollars per year) and a Mac computer, so it warrants a separate decision.

12. Troubleshooting: frequent questions

Symptom	Cause and solution
The Install button does not appear on the phone.	On Android only Chrome offers it, on iPhone only Safari. Check that the address is https://YOURNAME.github.io and the file was not opened directly.
The application does not update to the new version.	Close the application completely, then open it again with an internet connection. The update activates on the second opening.
The medical report does not open.	The report opens on a new page: allow pop-ups for this address in the browser settings, then press Create report again.
The application questions the value when saving.	This is the typo protection: it asks for confirmation before values below 2.0 mmol/l or above 25.0 mmol/l. A real extreme value can be saved with Yes.
An “unverified app” screen appears at the Drive connection.	This only appears if the Google Cloud project is still in Testing status — in that case proceed with Continue, but only registered test users have access. In production status (see chapter 9.2) this screen does not appear.
The follower cannot find the shared file in the picker.	Check that the folder on Drive is shared with the follower’s exact e-mail address and that the owner’s application has synced at least once (that is when the file is created).
No alert arrives on the follower’s phone.	In the Alert settings the Enable notifications button must be green, and the application must be open or running in the background. With the application fully closed the alert does not arrive.
Drive asks to sign in again every hour.	By Google’s security rule, access renews roughly every hour. This usually happens by itself, without a pop-up window.
The CGM file import reports an error.	Check that the original, unmodified CSV file from LibreView or Clarity is being loaded. A file opened and re-saved in Excel may have a changed structure.
Data disappeared after cleaning the browser.	The application’s dual storage (backup database) usually restores it automatically on restart. If not, everything can be reloaded from the latest JSON backup with the Import button.
A few Hungarian words remain in English mode.	Manually entered data (food names, notes, insulin names) intentionally stays exactly as recorded — that is the user’s personal data.
The Sync now button shows an error message (for example “Login expired”, “No permission”, “File not found”, “No internet connection”).	This precise error message (introduced in v11.1) helps pinpoint the problem: for an expired login, sign in again with Connect with Google account; for a missing permission, check that the folder is shared; for a deleted file, choose the folder again; without an internet connection, the data syncs on the next successful attempt. The scheduled, automatic sync retries silently in this case, without showing an error.

13. Medical and privacy notes

13.1 From a specialist's point of view

The application's dose suggestion follows the internationally common three-part formula: carbohydrate coverage plus correction minus insulin on board (IOB); every parameter is entered on the Settings tab under medical supervision. The detailed, step-by-step derivation and worked examples of the formula are given in Chapter 6 (Bolus calculation in detail). The estimated HbA1c is computed from the average glucose of the selected period with the common conversion formula; alongside it, v9.0 also calculates the GMI (Glucose Management Indicator) with the international formula. The application consistently marks both as estimates. The medical report provides a summary of any period that can be printed or forwarded as a PDF, with SD/CV variability measures. Follower mode is read-only: the treating physician sees the patient's data but cannot modify it.

What to review during the appointment: a checklist

The following list gives a fast, factual overview from the application at the quarterly check-up, illustrated with examples.

- Medical report for a 90-day period: every key figure and the detailed log on a single printable page — the fastest starting point.
- Statistics tab, 90-day range: TIR, average, estimated HbA1c and GMI together. Example: with 76% TIR and 7.1% estimated HbA1c the direction of therapy is good; details can be refined on the charts.
- Daily pattern chart: the hourly averages. Example: if the 3–6 am band is repeatedly low, lowering the basal dose may come up — a decision for the specialist.
- Measurement summary: the ratio of low, normal and high readings. Example: 3 low, 41 normal, 9 high indicates that the excursions tend upward.
- Carb/insulin ratio card: the actual ratio computed from the diary against the configured ICR. Example: if the configured morning ICR is 10 but the diary shows 1 unit covering only 8 grams, adjusting the ratio may be justified.
- Entries tab with the notes: the context of restaurant, sport and stressful days, as in the examples of chapter 7.

13.2 Where the data is

Every entry is stored on the given device, with dual storage: besides the fast primary store, a backup database also keeps it, so it recovers even after an accidental browser clean-up. The application sends no data whatsoever to the developer or any third party, and contains no advertising or tracking code. With Drive sync enabled, a copy of the diary also exists on the user's own Google Drive, and only those with whom the user themselves shared the folder can access it. Sharing can be revoked at any time on the Drive interface, and sync can be switched off with the Disconnect button. The medical report is generated locally, on the device — its content reaches anyone only if the user prints or sends it themselves.

13.3 Summary

The HBC Diabetes Diary v19 Type 1 Diabetes APP is an installable, offline-capable, bilingual diary that turns logging into calculation, feedback and sharing: it suggests insulin doses based on values agreed with the physician, warns at low and unusual values, shows the trends on charts, statistics and the daily pattern, produces a printable medical report for any period, and can be followed live on the designated follower's phone. Installation consists of a one-time publication and a few taps per device; Drive sync works by itself after a one-time Google setup. The chapters of this document walk through every phase from everyday use to the store release, and chapter 12 resolves the most frequent difficulties.

14. The SOS emergency feature

New in v12.0. During severe hypoglycaemia an insulin-treated person may become confused or lose consciousness — and then every minute counts for unknown helpers and paramedics. The SOS page shows the essentials in large letters with a single tap: what happened, what to do and whom to call.

14.1 Setting it up

Open Settings → Emergency (SOS) and fill in the following. All data is stored only on your device.

- **Full name and own phone number:** these also appear in the header of the Medical report, making it an official, identifiable document.
- **Home address:** to inform helpers and paramedics.
- **Contacts to notify:** any number of people can be added with the + (add) button (name, relation, phone number); the first person in the list is the primary contact. Any entry can be removed with the bin icon.
- **"How to communicate with me" note:** a fully, freely editable text. Based on your own experience, describe what you are like during hypoglycaemia (calm? confused? dismissive? — often as if you were drunk) and how helpers should treat you while you are still conscious. Helpers will see this text word for word.

14.2 Opening the SOS page

- **On a phone:** with the red SOS button at the bottom of the  hamburger menu in the header.
- **On a desktop computer:** with the small button on the right of the header (mainly useful for checking the settings).

The page appears in the configured language (Hungarian or English) and contains: the warning title ("INSULIN-TREATED DIABETIC"), step-by-step first-aid instructions, the 112 emergency call button, one-tap call buttons for the contacts, the home address and the communication note.

From v12.2 the tiles can be freely re-arranged by dragging the  handle in their top right corner — the order is remembered on the device. In the default order the Contacts to notify tile sits above the 112 button. The note title also shows the nickname configured in Settings ("How to communicate with me — my name is ..."), and the page keeps the same, highly readable colours in dark mode too.

14.3 Automatic alert on a low reading

If the most recent reading of the day — a manual entry or an imported CGM value — falls below the configured low blood-glucose threshold, the SOS page appears automatically with the measured value, and (if notifications are enabled) a system notification is also sent. It is dismissed with the "I AM OK — dismiss alert" button (in follower mode the button reads "OK — dismiss alert"); the application alerts only once per reading, and an older (retrospectively recorded) value never triggers an alert.

14.4 Important limitations and advice

- **It cannot appear on a locked screen:** a web application cannot draw over the lock screen of a phone and cannot unlock the phone — this is a system-level restriction that applies to every application.
- **Also set up the system-level emergency data:** the "Medical ID" of the Health app on iPhone, or "Emergency information" on Android — these are available WITHOUT unlocking, from the lock screen, and paramedics look there first. The SOS page complements them; it does not replace them.
- **Keep the app on the home screen:** after unlocking the phone the helper finds it immediately and can open the SOS page from the menu with a single tap.
- **Not a medical device:** the SOS page is an informational aid; in the case of severe illness always call an ambulance.